A Unified Remote Sensing Object Detector Based on Fourier Contour Parametric Learning

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***Abstract*— A unified object detector needs to integrate various abilities for adapting to different remote sensing object detection tasks. However, there is a lack of a feasible way to integrate multigrained object detection requirements i.e., horizontal bounding box (HBB), oriented bounding box (OBB), and instance segmentation (InSeg) into a unified detection way. Then, it often has to design specific parametric learning ways and their corresponding architectures, which cannot be finely adaptive to various kinds of object detection tasks. Therefore, in this article, a new benchmark is set up to integrate multigrained object detection requirements of HBB, OBB, and InSeg into one challenging task of arbitrary-shaped object contour detection. At the same time, a unified object contour detector (UniconDet) is proposed for achieving multigrained object detection from complicated remote sensing scenes. First, a Fourier contour parametric modeling (FCPM) is defined to project arbitrary-shaped object contours from the spatial domain into the frequency domain. Then, it can unify spatial parametric representations of HBB, OBB, and InSeg as frequency coefficient representations, which can be used for realizing a more generic and robust parametric regression. Second, a multiview cross-attention (MVCA) feature extraction way is designed at each scale of the regression layer, which can assist UniconDet in perceiving Fourier contour parameters by exploring the coupled relations between different discrete contour sampling periods of each object. Third, a center-contour enhancing regression layer (C^2-ERL) is designed to generate regional guidance and cascade contour propagation, which can ensure a more accurate center point prediction and Fourier contour parameter regression. Finally, extensive experiments are carried out on benchmarks of HBB, OBB, InSeg, and new multigrained object detection, and the results indicate that our proposed UniconDet can obtain superior performance. The source code is available at https://github.com/ZhAnGToNG1/UniconDet.**

***Index Terms*— Fourier contour parametric modeling (FCPM), multiview cross-attention (MVCA), remote sensing object detection, unified object detector.**

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I. Introduction

CURRENTLY, there exist three different tasks for remote sensing object detection, including the horizontal bounding box (HBB), oriented bounding box (OBB), and instance segmentation (InSeg), which aim to accurately localize different category objects from complex remote sensing scenes. Although massive efforts have been made for HBB [1], [2], OBB [3], [4], and InSeg [5], [6] in the remote sensing domain, they cannot also satisfy multigrained object detection requirements for different categories of remote sensing objects. As shown in Fig. 1(a), in a very complicated airport (AT) scene, the OBB can be adopted for the detection of rectangular cars. However, since the storage tanks (STs) do not have apparent orientation information, the OBB is not suitable for their detection. Moreover, because of the irregular shape of airplanes (ALs), neither OBB nor HBB are very viable options for AL detection. Therefore, several studies [7], [8], [9], [10], [11], [12], [13] attempt to establish a unified remote sensing object detector to integrate the multigrained detection abilities involving HBB, OBB, and InSeg. Xu et al. [7], Shi and Zhang [8], and Liu et al. [9] are all considering utilizing a multitask learning strategy to integrate the InSeg ability with HBB or OBB for improving object detection ability based on parallel, cascaded and shared feature extraction architecture. In order to set up the multigrained object detection ability, Yang et al. [10] and Qian et al. [11], respectively, designed their unified detectors for integrating HBB and OBB detection abilities by HBB and OBB conversion according to a specific weakly self-supervised learning way and horizontal smallest enclosing rectangle constraint. Besides, based on the segment anything model (SAM), Chen et al. [12] employed various prompting mechanisms (i.e., the bounding boxes and queries) to integrate HBB and InSeg detection abilities into a generic framework called RSPrompter. Zhang et al. [13] considered HBB and OBB detection tasks as universal language modeling and designed a visual-text alignment representation learning to integrate HBB and OBB detection abilities into a multimodal large language model called EarthGPT. Nevertheless, whether multitask learning, different detection granularity conversion, prompt engineering, or universal language modeling, they are all dependent on constructing multigrained labeled datasets, designing valid strategies for fundamental model integration and undergoing expensive training procedures. Then, through tedious dataset preparation, specific model design, and lengthy model training procedure, these studies [7], [8], [9], [10], [11], [12], [13] also cannot well integrate multigrained object

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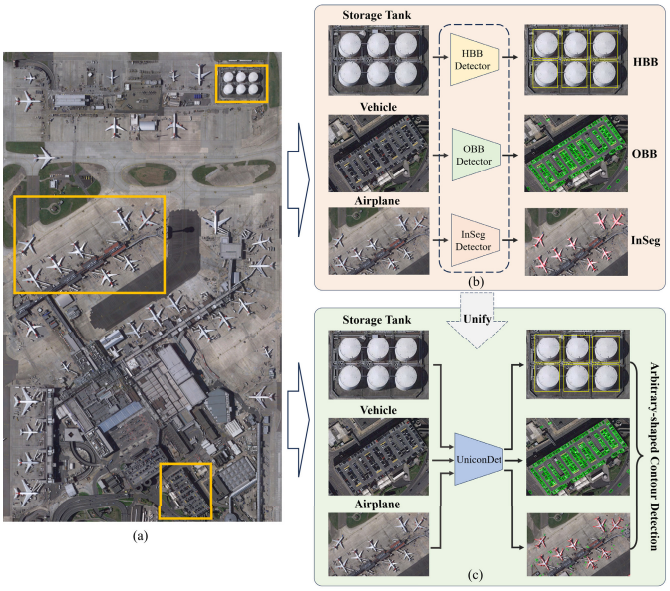


Fig. 1. Illustration of the multigrained object detection requirement in the complex remote sensing scene. (a) Complex AT scene including multiple categories, e.g., AL, V, and ST. (b) Individual detectors are designed for different categories. (c) Proposed UniconDet is applied for detecting all kinds of objects.

detection requirements of HBB, OBB, and InSeg. They can only unify HBB and OBB or HBB and InSeg for detection; thus, it would hinder precise and adaptive detections of multicategory objects from complicated remote sensing scenes. Sebsequently, how to integrate multigrained object detection abilities (i.e., HBB, OBB, and InSeg) into a unified detection way becomes a challenge, which has to be further studied for catering to different detection granularity of multicategory objects, while advancing the development of remote sensing object detection technique.

Considering the abovementioned issues, a new benchmark is set up to unify multigrained object detection requirements of HBB, OBB, and InSeg into a challenging task of arbitrary-shaped object contour detection, which can develop a versatile yet accurate object detection capability for different category objects, while addressing the limitations of currently existing benchmarks. Referring to the newly built benchmark, a novel unified object contour detector (UniconDet) is proposed to achieve multigrained object detection from complex remote sensing scenes, as described in Fig. 1(b) and (c), which can integrate the multigrained object detection abilities of HBB, OBB, and InSeg into one concise object detection framework according to Fourier parametric learning. First, a novel Fourier contour parametric modeling (FCPM) is defined for arbitrary-shaped object contour representations, which is shown in Fig. 2(a). It can project arbitrary-shaped object contours from the spatial domain into the frequency domain by a designed spatial object contour sampling strategy and Fourier transform, which can unify the spatial parametric representations of HBB, OBB, and InSeg as Fourier contour parameters in the frequency domain and be considered for parametric regression. Because of the scalable decomposition of high-low-frequency coefficients and feasible spatial-frequency conversion, the designed FCPM can provide more robust and stable regression performance than the currently existing spatial parametric representations. Second, to assist the proposed UniconDet in learning Fourier contour parameters, a new multiview cross-attention (MVCA) feature extraction approach is designed at each scale of the regression layers. It can help the model to perceive Fourier contour parameters by exploring the coupled relations between different discrete contour sampling periods of each object. As illustrated in Fig. 2(b), the spatial affine transformation is used for each scale of the regression layers, which can generate different transformed views for each feature map to perform the equivalent periodic shifts of discrete contour sampling. And it can be incorporated with the cross-attention mechanism to achieve alignment of discrete contour sampling for each object with different periods, which can effectively improve Fourier contour parameter perception ability. Third, followed by the more effective Fourier contour parameter perception, a center-contour enhancing regression layer (C^2-ERL) is designed to ensure Fourier contour parameter prediction including center points and Fourier degrees. As shown in Fig. 2(c), the regional guidance is designed in C^2-ERL to provide more descriptive information for center points. At the same time, the cascade contour propagation among all scale layers is designed to integrate multiscale information to effectively promote scale-invariant Fourier contour parameter prediction. Finally, extensive experiments are executed on three conventional detection benchmarks (e.g., HBB, OBB, and InSeg) and a newly set up multigrained object detection benchmark, and results indicate that the proposed UniconDet can achieve or approach the state-of-the-art (SOTA) performance on many benchmarks.

In general, the main contributions can be summarized in the following four points.

1) A new benchmark for multigrained object detection is constructed for developing a versatile yet accurate object detection capability in the remote sensing domain. Subsequently, two datasets of multigrained object detection (i.e., MG-DOTA and MG-NV10) are built for capability evaluation. Meanwhile, a unified remote sensing object contour detector called UniconDet is also proposed for arbitrary-shaped object contour detection, and thus, it not only can adapt to existing HBB, OBB, and InSeg object detection tasks, but also cater to the multigrained object detection requirements.
2) An FCPM is first defined to project the arbitrary-shaped remote sensing object contours from the spatial domain into the frequency domain, which can effectively unify spatial parametric representations of HBB, OBB, and InSeg as a single Fourier contour parameter representation, while achieving more robust and stable regression.
3) A novel feature extraction method of MVCA is designed within each scale of the regression layer for assisting in Fourier contour parameter perception. It can utilize the view transformation to perform the equivalent periodic shifts of discrete contour sampling. Meanwhile, the coupled relations among different discrete contour sampling periods of each object are explored by a cross-attention

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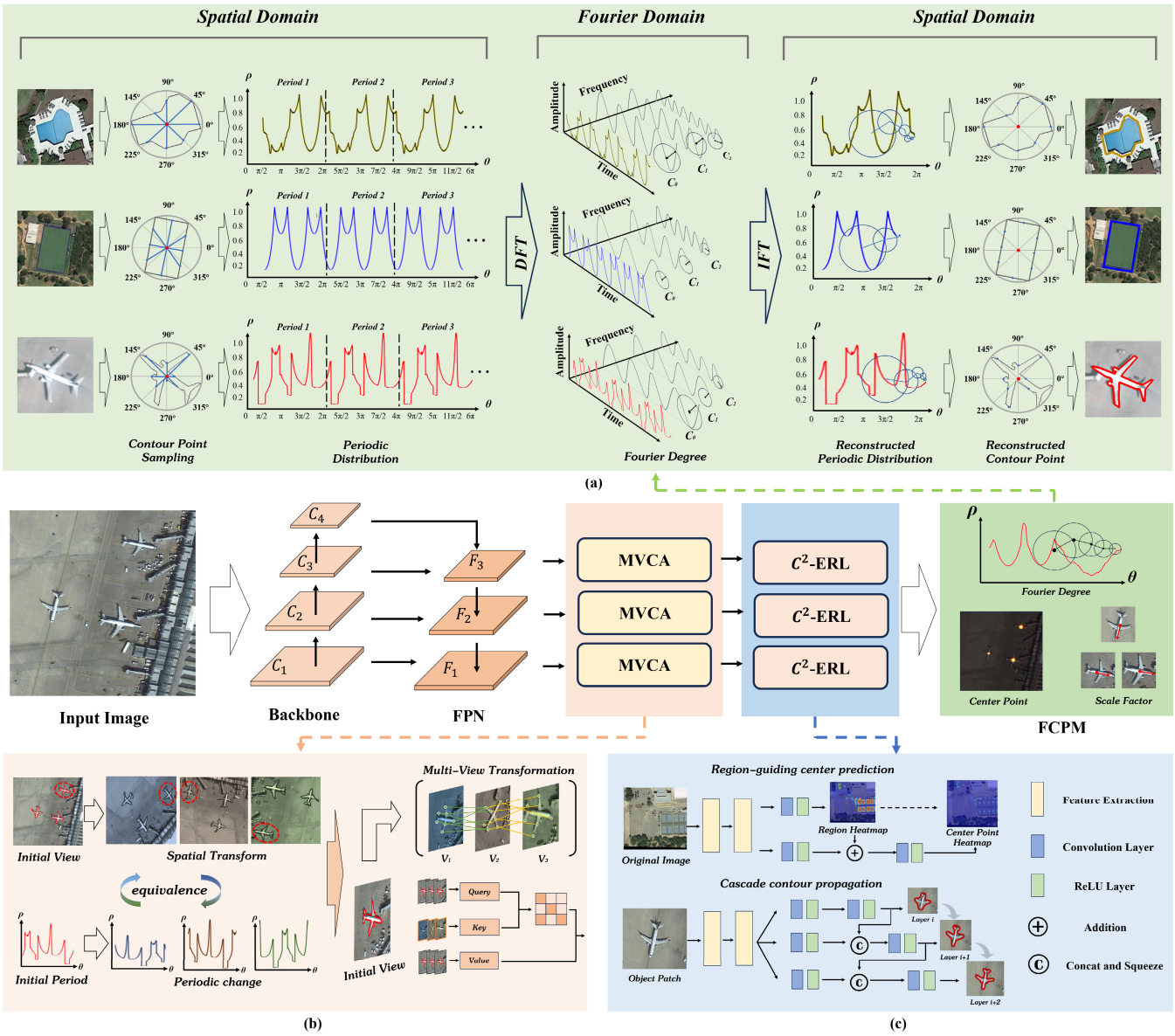


Fig. 2. Overall framework of the proposed UniconDet. (a) Modeling process of the proposed FCPM. (b) Designed MVCA. (c) C^2-ERL in the head network.

mechanism, which can effectively advance the Fourier contour learning.

4) According to the newly designed FCPM and MVCA, a novel C^2-ERL is designed for ensuring accurate Fourier contour parameter prediction, which can provide the useful regional guidance and cascade contour propagation to jointly learn the Fourier contour parameters for center point prediction and Fourier degree regression.

II. Related Work

A. Unified Object Detector

Detection and segmentation tasks have been long-standing challenges in the computer vision field, which require understanding what and where the interested objects are. Along with the demand of different tasks, specialized designs are tailored to provide the needed information. Unfortunately, customized designs for different tasks not only disperse research efforts, but also hamper the practical applications of algorithms. To this end, some progress has been made toward achieving a universal architecture design for multiple tasks. For example, image segmentation tasks are often categorized into semantic, instance, and panoptic segmentation in the natural scene. To train all of the tasks on one framework, K-Net [14], Mask2Former [15], OneFormer [16], and FreeSeg [17] have been proposed to unify these independent segmentation tasks. In addition, there are some works attempting to set up a unified architecture for object and segmentation tasks. For instance, Li et al. [18] proposed a unified framework, which extends the object detector called DINO by adding a mask prediction branch that supports all image segmentation tasks. Moreover, the latest trends involving the universal design have raised in numerous object detection tasks. Specifically, in the 3-D object detection task, due to the substantial differences in object

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distribution and point density within point clouds collected from various environments, some works [19], [20] have made efforts to address indoor and outdoor 3-D detection within the same framework. In the open-vocabulary detection task, many researchers [21], [22] have begun to utilize a pretrained text encoder to encode all kinds of visual concepts and learn a common semantic space for them. Regarding the camouflaged object detection task, Dong et al. [23] proposed a unified framework based on uncertainty-aware feature reassembly for diverse binary segmentation tasks. In addition to the universal architecture design for natural scenes, Li et al. [24] proposed a novel grid-level mixture of experts (MoE) approach and a dynamic learning rate adjustment strategy to achieve the unification of the tasks from different modalities, i.e., optical and synthetic aperture radar (SAR). However, given the fragmented research status across tasks such as HBB, OBB, and InSeg, it is necessary to unify these tasks and promote the development of remote sensing object detection.

B. Contour Parametric Modeling

The contour parametric modeling has been employed on text detection [25], [26], [27], [28], [29], [30] and InSeg [31], [32], [33], [34] tasks. Compared with pixel-level classification modeling methods, the contour-based methods can directly obtain the contour points of arbitrary-shaped objects, as follows:

$$(x_i, y_i), \quad i \in P \tag{1}$$

where x and y denote the spatial coordinates of each contour point, and P represents the set of all contour points. In general, these contour points can be sampled at equal angles around the center point or centroid of the circle. Early on, many works [32], [34], [35], [36] adopted the multistage process, aiming at iteratively optimizing the manually designed initial contours to fit each object boundary. For example, Ling et al. [35] evolved an initial contour to match the object boundary by a graph convolutional network (GCN). And then, Peng et al. [34] introduced the circular convolution for deforming an initial contour by regressing vertexwise offsets. However, the handcrafted initial contour exists huge differences and uncertainty with the actual object contour, which would lead to invalid vertex pairs and confuse the training process. To avoid the issue, Zhang et al. [32] adopted a learnable contour initialization strategy, which specifies the shape of the initial contour by the network. Meanwhile, Xie et al. [31] also directly predicted the contour of instances through instance center classification and dense distance regression in a polar coordinate. In addition, some works considered that discrete contour sampling is not efficient. Recently, a number of researches began to encode the object contour into a shape vector. For instance, Xu et al. [33] proposed the explicit shape representation to decode the object contour points with tensor operations, which largely reduces the computation. For the same reason, some works about text detection [25], [27], [28] also focused on the shape encoding. Wang et al. [27] mapped the complex text contour discovery problem from shape space to parameter space by using the Chebyshev approximation to fit the function curve. Similarly, discrete cosine transform (DCT) [28] and Fourier transform [25] were also used to encode the discrete contour points. By comparison, the contour parametric modeling from a general modeling view not only can cope with mass contour points but also inherit the advantages of the one-stage paradigm.

As a result, the contour parametric modeling is promising for achieving unified parametric modeling for multigrained object detection requirements. However, there is little research [37], [38] focusing on applying the contour parametric modeling to remote sensing imagery (RSI) because complex object characteristics and background noise hinder the development of contour parametric modeling. Thus, it is necessary to explore the potential of contour parametric modeling to achieve arbitrary-shaped contour detection for objects of different categories in RSI.

C. Attention Mechanism in Object Detection

Regarding the development of attention mechanisms in the field of object detection, it can be categorized into two phases. Early on, attention mechanisms were applied to convolutional neural network (CNN) architecture to improve the detection performance. For example, Dong et al. [39] proposed attention-based multilevel feature fusion modules to integrate the multilevel outputs of feature pyramid networks (FPNs) adaptively, enabling multiscale object detection. In addition, Zhou et al. [40] proposed an attention-guided densely multiscale network to detect salient objects, which attempts to enlarge the receptive field to locate salient objects by generating the soft spatial attention map. And then, to tackle the issue of few-shot object detection, Xin et al. [41] proposed the extensible attention mechanism to assist the few-shot detector in inferring co-existing unseen features according to the provided local part. Afterward, the advent of ViT [42] has ushered in a new era of attention mechanisms, and its self-attention and cross-attention mechanisms provide a powerful way to model long-range and cross-modal dependencies. Built on the ViT, DETR [43] initially broke away from conventional anchor-based approaches, leveraging global self-attention to cast the object detection task as a set prediction problem. However, Dai et al. [44] considered the complexity of self-attention to be quadratic with respect to the input spatial scales, which limits the resolution of input feature maps and makes extracting features infeasible. To overcome the issues, they applied dynamic attention on full scales of feature pyramid based on the spatial, scale, and representation importance. In addition, Zheng et al. [45] thought that treating all tokens in ViT [42] equally without discrimination would lead to a redundant computational burden. Thereby, they also proposed a Focus-DETR, which focuses attention on more informative tokens and abandons the background queries. In the few-shot object detection, Han et al. [46] proposed the asymmetric-batched cross-attention to aggregate the key information from the two branches (i.e., query image and support image branches) with different batch sizes, which can promote the multilevel interactions between the query and support inputs.

In general, both the self-attention and cross-attention mechanisms enable the establishment of interactive relationships

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among different inputs. Motivated by this, we make use of the attention mechanism to construct a feature extraction method, aiming to achieve effective object contour perception.

III. Methodology

In this section, a unified parametric modeling FCPM is illustrated in Section III-A. Moreover, a novel feature extraction method of MVCA is introduced in Section III-B. Then, a C^2-ERL is elaborated in Section III-C. Finally, the details of the loss function are presented in Section III-D.

A. Fourier Contour Parametric Modeling

As mentioned above, a new benchmark is set up to integrate currently existing remote sensing object detection tasks (i.e., HBB, OBB, and InSeg) into a challenging task of arbitrary-shaped object contour detection. Then, referring to the newly built benchmark, a unified contour-based parametric modeling method of FCPM is designed to project arbitrary-shaped object contours from the spatial domain into the frequency domain, which can unify the spatial parametric representations of HBB, OBB, and InSeg as a generic representation of Fourier contour parameters in the frequency domain, while setting up a feasible spatial-frequency conversion way based on the designed spatial object contour sampling strategy and Fourier transforms. Subsequently, a series of Fourier contour parameters can be used to replace commonly adopted spatial parametric representations for more robust and stable regression, thus adaptively catering to multi-grained object detection requirements of different category remote sensing objects. Next, more detailed definitions of FCPM are elaborated as follows.

Specifically, as illustrated in Fig. 2(a), the polar coordinate system is first constructed for representing arbitrary-shaped object contours as the combinations of center points and their corresponding discrete contour sampling points. When considering center points as the poles, several discrete contour points can be obtained by anticlockwise sampling with equal angular intervals from the rightmost end, such as $\theta = [\theta_1, \theta_2, \ldots, \theta_N]$, where $\theta_n = n \times (2\pi/N)$, $n \in [1, N]$, and N denotes the number of discrete contour sampling points. These rays are emitted from the pole and intersect with the contour edges, in which these line segments can be expressed as $\rho = [\rho_1, \rho_2, \ldots, \rho_N]$. Hence, these polar coordinates can be represented as $[\rho_n, \theta_n]$. Besides, to eliminate the impact of large-scale variances on parametric modeling, a series of line segments is normalized by the scale factor, i.e., $\rho_{\max} = \text{Max}(\rho_1, \rho_2, \ldots, \rho_N)$. This operation can be calculated by $\rho_m = [\rho_1^m, \rho_2^m, \ldots, \rho_n^m] = [\rho_1, \rho_2, \ldots, \rho_N]/\rho_{\max}$. Subsequently, as illustrated in Fig. 3(b) and (c), it is evident that the normalized polar coordinates (ρ_n^m, θ_n) no longer have huge scale variances between different scale objects, which is beneficial for the equal optimization process of each category object contour. Formally, (ρ_n^m, θ_n) can be formulated into the complex value, i.e., $f(n/N) = (\rho_n^m, \theta_n)|_{n=1}^N = \rho^m(n) + i\theta(n)$. Then, the discrete Fourier transform (DFT) is further applied on the periodic distribution $f(n/N)$ to produce a series of Fourier degrees c_k in the frequency domain, i.e., Fourier domain, as presented in the following equation:

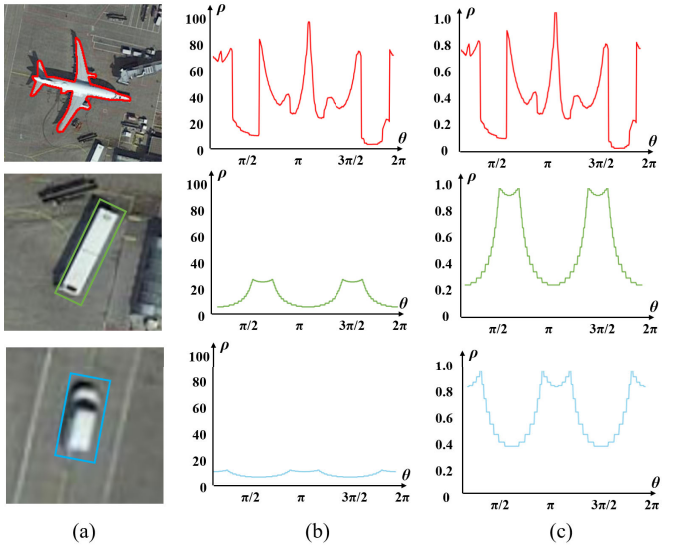


Fig. 3. Discussion about the curve changes of contour sampling points in polar coordinate system. (a) Instances with different scales. (b) Curves of original contour sampling points. (c) Curves of normalized contour sampling points. Here, the ρ and θ represent the polar diameter and angle in the polar coordinate, respectively.

$$c_k = \frac{1}{N}\sum_{n=1}^{N} f\left(\frac{n}{N}\right) e^{-2\pi i k \frac{n}{N}} \tag{2}$$

where $k \in [1, K]$ represents the frequency component, and c_k denotes a series of Fourier degrees, which are used for characterizing the initial state of frequency k and its specific form $c_k = u_k + iv_k$, where u_k and v_k are the real and image parts of a complex number. Commonly, K is empirically set as 10. Then, the contour can be represented in the frequency domain as a compact $2(2K+1)$-dimensional Fourier degree $[u_{-K}, v_{-K}, \ldots, u_0, v_0, \ldots, u_K, v_K]$. Among them, the number 1 in $2(2K+1)$ denotes the Fourier degree of zero-frequency component c_0. During the training phase, the objective is to encourage the proposed UniconDet to well learn $2(2K+1)$-dimensional Fourier degrees for arbitrary-shaped object contour detection.

Then, at the inference phase, these predicted Fourier degrees are first converted as the scale-normalized contour points by inverse Fourier transform, which can be expressed as follows:

$$f\left(\frac{n}{N}\right) = \sum_{k=-K}^{K} c_k e^{2\pi i k \frac{n}{N}}. \tag{3}$$

In (3), each component $c_k e^{2\pi i k(n/N)}$ represents a circular motion with fixed frequency component k. It can be found in Fig. 2(a) that the larger circles represent lower frequency components to depict the coarser-grained contour information; otherwise, the small circles indicate high-frequency components to depict the fine-grained contour information. After that, the contour points in the polar coordinate system are restored by multiplying the predicted normalized factor $\rho_{\max}$. Lastly, combined with the predicted center point, the original

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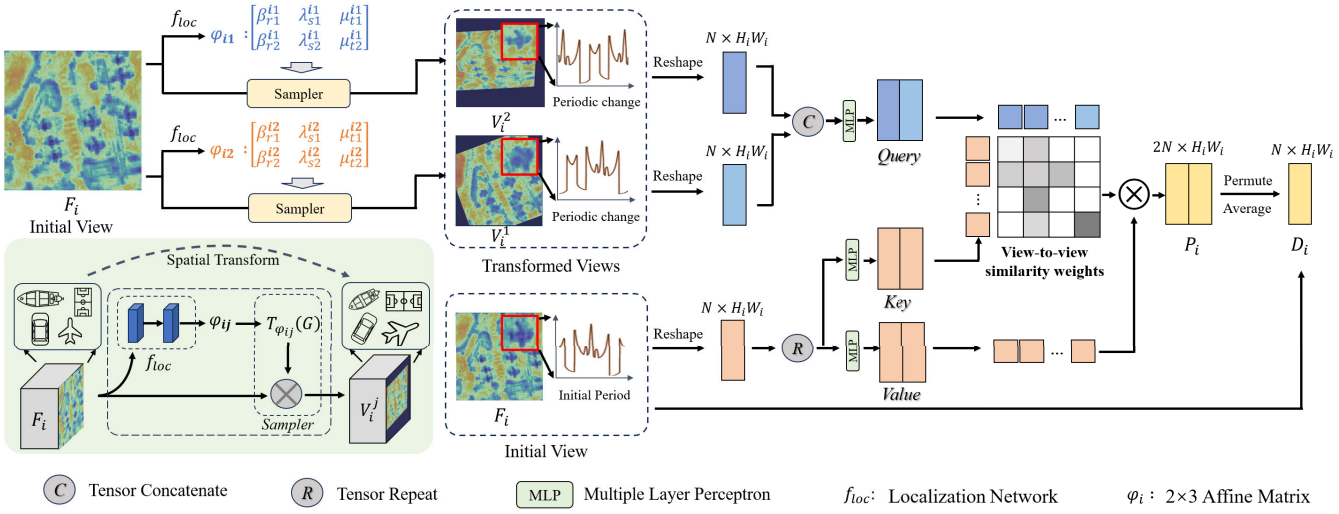


Fig. 4. Illustration of the proposed MVCA. First of all, different views are generated by the spatial transformer in the light green area. And then, a view-to-view cross-attention mechanism with query–key–value embeddings is utilized for interacting with them.

contour points (x_n, y_n) in the image can be calculated by the following equation:

$$(x_n, y_n) = (\rho_n \times \cos(\theta_n) + x_{ct}, \rho_n \times \sin(\theta_n) + y_{ct}) \tag{4}$$

where x_{ct} and y_{ct} represent the spatial coordinates of predicted center points. Obviously, arbitrary-shaped object contour can be described by the newly defined FCPM. Then, because of the scalable decomposition of high- and low-frequency components and feasible spatial-frequency conversion, the defined FCPM can provide a more robust and stable regression performance compared to utilizing the currently existing spatial parametric representations.

B. Multiview Cross-Attention

Referring to the newly defined FCPM and aiming to assist UniconDet in learning Fourier contour parameters, as shown in Fig. 2(b), a feature extraction way of MVCA is devised to advance Fourier contour parameter perception by exploring the coupled relations among different discrete contour sampling periods of every object at each scale layer.

Specifically, the FPN is adopted to integrate different scale features from the backbone, aiming to acquire multiscale feature maps $F_i \in \mathbb{R}^{B\times N\times H_i\times W_i}$, $i \in [0, 2]$ for arbitrary-shaped object contour detection, where B and N denote the batch size and channel dimensions, respectively, and H_i and W_i indicate the height and width of the ith feature map respectively. Then, as illustrated in Fig. 4, the spatial transformer network [47] is applied to obtain different views of feature maps at each scale layer. Thus, a 2×3 affine matrix φ_{ij} is generated by feeding the F_i into a learnable localization network f_{loc} as follows:

$$\varphi_{ij} = f_{\text{loc}}(F_i) = \begin{bmatrix} \beta_{r1}^{ij} & \lambda_{s1}^{ij} & \mu_{t1}^{ij} \\ \beta_{r2}^{ij} & \lambda_{s2}^{ij} & \mu_{t2}^{ij} \end{bmatrix}. \tag{5}$$

In (5), f_{loc} is composed of two 3×3 convolution layers with rectified linear unit (ReLU) activation functions and one adaptive average pooling layer; β_{r1}, β_{r2} and λ_{s1}, λ_{s2} represent the coefficients of rotation and scaling; and μ_{t1} and μ_{t2} denote the coefficients of translation. Next, the φ_{ij} is adopted for affine transformation T. In detail, the coordinates $G = (x_i^t, y_i^t)$ of the output feature map are multiplied with the φ_{ij} to map the coordinate (x_i^s, y_i^s) of the input feature map F_i, as follows:

$$\begin{pmatrix} x_i^s \\ y_i^s \end{pmatrix} = T_{\varphi_{ij}}(G) = \begin{bmatrix} \beta_{r1}^{ij} & \lambda_{s1}^{ij} & \mu_{t1}^{ij} \\ \beta_{r2}^{ij} & \lambda_{s2}^{ij} & \mu_{t2}^{ij} \end{bmatrix} \begin{pmatrix} x_i^t \\ y_i^t \\ 1 \end{pmatrix}. \tag{6}$$

In (6), since the float coordinate values are usually presented in (x_i^s, y_i^s), the bilinear interpolation sampling strategy is used to get the integer values and fill them into pixel values of (x_i^t, y_i^t) for the output feature map with a different view. Subsequently, to repeat the abovementioned process, each scale feature map with different transformed views $V_i^j \in \mathbb{R}^{B\times N\times H_i\times W_i}$ would be generated. Here, the number of transformed views M_v is empirically set as 2, i.e., $j \in [0, M_v - 1]$. At each scale of the regression layer, every object in different transformed views of the feature map can be seen as the equivalent periodic shifts of discrete contour sampling according to FCPM. Afterward, an initial feature map F_i and multiview feature maps V_i^j can be further interacted with each other in the form of cross-attention based on query-key-value embeddings, as illustrated in Fig. 4. The purpose is to adopt a cross-attention mechanism for capturing long-range dependencies and then facilitating the perception of Fourier contour parameters, which would effectively explore the coupled relations among different discrete sampling periods of each object contour. Accordingly, before setting up query–key–value embeddings for cross-attention, the initial feature F_i and generated multiview feature maps V_i^j are converted into the same dimension, i.e., $\mathbb{R}^{B\times N\times H_i W_i}$. Then, as shown in Fig. 4, the key and value are both acquired from the initial feature F_i, and the query can be obtained from the generated multiview feature maps V_i^j, which can be expressed as

$$\begin{aligned} k_i, v_i &= f_{k,v}(\text{Repeat}(F_i, M_v)), \quad k_i, v_i \in \mathbb{R}^{B\times M_v N\times W_i H_i} \\ q_i &= f_q\left(\text{Concat}\left(V_i^0, V_i^1, \ldots, V_i^j\right)\right) \end{aligned} \tag{7}$$

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$$q_i \in \mathbb{R}^{B\times M_v N\times W_i H_i}. \tag{8}$$

In (7) and (8), Repeat(·) is a repetitive operation, which can copy the initial feature F_i for M_v times to generate the key k_i and value v_i according to the learnable weighting function $f_{k,v}(\cdot)$. In addition, Concat(·) indicates the feature concatenation, and then, it can be adopted to integrate different transferred views of the initial feature map to produce query q_i via the learnable weighting function $f_q(\cdot)$. Then, in Fig. 4, the view-to-view similarity weights W_i can be calculated by

$$W_i = \text{Softmax}\left(q_i {k_i}^{\mathrm{T}}\right) W_i \in \mathbb{R}^{B\times W_i H_i\times W_i H_i}. \tag{9}$$

In (9), W_i represents the coupled relations between the initial feature map and its corresponding different transformed views, which can be calculated by conducting the dot-product operation between q_i and k_i. Subsequently, view-to-view similarity weights W_i would be multiplied with the initial feature map of value v_i and then produce contour perceptive feature D_i, which can be expressed as follows:

$$P_i = W_i v_i, \quad P_i \in \mathbb{R}^{B\times M_v N\times W_i H_i} \tag{10}$$

$$D_i = \left(\sum_{k=0}^{M_v} P_i^k\right) \Big/ M_v, \quad P_i^k, D_i \in \mathbb{R}^{B\times N\times W_i H_i}. \tag{11}$$

In (10) and (11), the coupled relations W_i among different transformed views would be fused into P_i, and then, the contour perceptive feature D_i can be obtained by averaging the P_i across the dimension of views. Particularly, to mitigate the latent impact on model convergence, a residual connection structure is adopted for the output feature D_i and the initial input feature F_i. After (10) and (11), the designed feature extraction method of MVCA can be achieved at each scale of the regression layer, and then, it can assist UniconDet in perceiving Fourier contour parameters of each object from complex remote sensing scene.

C. Center-Contour Enhancing Regression Layer

Under the unified parametric modeling strategy of FCPM, and based on the effective feature extraction way of MVCA, it is essential to further guarantee the more accurate center point prediction and Fourier contour parameter regression of certain category objects. Thus, a novel C^2-ERL is designed to provide useful regional guidance and cascade contour propagation, as shown in Fig. 2(c). Subsequently, more details about the regional guidance and cascade contour propagation would be elaborated as follows.

First, regarding object center point prediction, and drawing inspiration from our previous study [48], a region-level instance prediction branch is designed to construct posterior guidance to facilitate center point prediction. Specifically, some objects in remote sensing scenes usually present unclear appearances, resulting in poor geometric characteristics, which would affect the performance of center point prediction, especially for tiny objects. Accordingly, extra contextual information is necessary to be captured to support object center point prediction from complicated remote sensing scenes. As the orange box shown in Fig. 5(a), a region-level instance prediction branch has the same structure as the center point

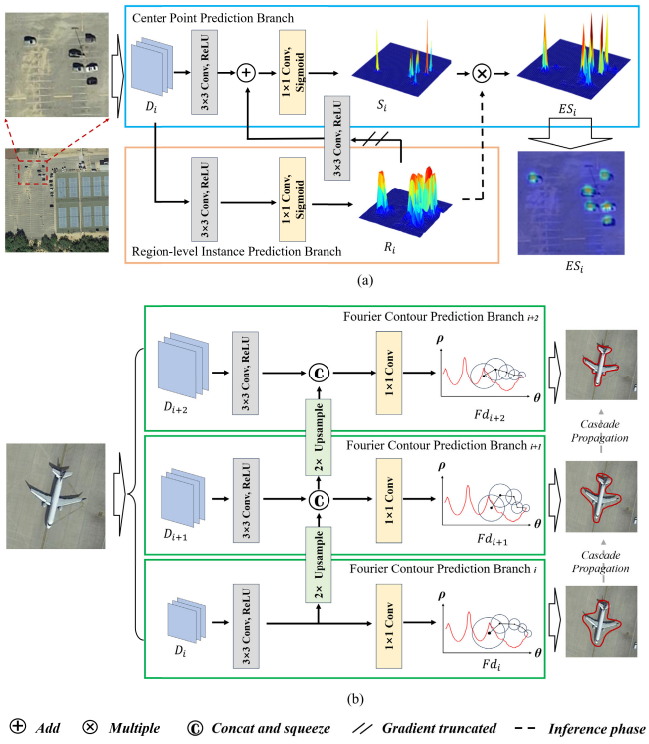


Fig. 5. Illustration of the proposed C^2-ERL. (a) Region-guiding center prediction. (b) Cascade contour propagation.

prediction branch, consisting of the cascaded 3×3 and 1×1 convolutions with the ReLU and Sigmoid activation functions. Notably, different from the center point prediction branch, the purpose of the region-level instance prediction branch is to produce object instances R_i based on instance supervised learning, which can collect more object-related contextual information to enhance object center point prediction in the aspects of model training and inference. At the model training phase, when the learnable R_i is considered to be injected into the intermediate process of feature extraction in the center point prediction branch, the gradient truncation is used to avoid the mutual influence between the two different tasks. Subsequently, a 3×3 convolution with the ReLU activation function is applied for R_i, and it can align the feature dimension to make R_i be injected into the center point prediction branch and enhance center point learning of S_i. Besides, according to the inference phase, as shown in the dashed line of Fig. 5(a), the region-level instance prediction R_i is multiplied by the prediction of center points S_i to get the ES_i along channelwise, which can further improve the center point response within each object region and then be beneficial to distinguishing tiny objects at the inference phase.

Second, regarding Fourier contour parameter regression, a cascaded contour propagation layer is implemented to transmit the mapping features that are used to regress contour parameters among different scale regression layers. This rule allows the model to progressively integrate the contour information from different scale layers, thereby promoting the regression performance of Fourier contour parameters. Specifically, since the Fourier degrees have been converted based on normalized contour sampling points, there is the

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same learning objective at different regression layers. Consequently, for relatively high-resolution scale layers, it is reasonable for them to adopt the prior knowledge from the previous low-resolution scale layer to assist in Fourier degree regression. In detail, the mapping feature of the low-resolution regression layer is upsampled into the same size as the adjacent higher resolution regression layer, and the concatenate-then-squeeze operation is performed on them, as shown in Fig. 5(b). After a 1×1 convolution, the cascaded and fused mapping feature is adopted for Fourier degree regression according to the higher resolution regression layer Fd_{i+1}. Then, following the above process, three layers of cascaded contour propagation are set up to ensure the accurate regression of Fourier degrees $\mathrm{Fd}_i, i \in [0, 2]$.

D. Loss Function

Regarding the overall loss L for model training, it consists of four parts, including the loss of the center point prediction L_{ct}, the loss of Fourier contour parameter regression L_{fd}, the loss of the scale factor prediction L_ρ, and the loss of regional guidance generation L_{reg}, which would be integrated into one equation in the following:

$$L = L_{ct} + L_{fd} + L_\rho + L_{\text{reg}}. \tag{12}$$

In (12), the keypoint learning way of CenterNet [49] is adopted for L_{ct}. Therefore, a size-adaptive Gaussian kernel $Y_{xyc} = \exp(-(((x - x_g)^2 + (y - y_g)^2)/2\sigma_p^2))$ is applied to smooth positive and negative samples for learning each center point (x_g, y_g) of ground truth. Here, $\sigma_p = ((2 \times \max(w, h) + 1)/6)$. Moreover, the focal loss [50] is also employed for L_{ct}, which can be expressed as follows:

$$L_{ct} = \frac{-1}{N_c} \sum_{xyc} \begin{cases} \left(1 - \hat{Y}_{xyc}\right)^\alpha \log\left(\hat{Y}_{xyc}\right) & \text{if } Y_{xyc} = 1 \\ \left(1 - Y_{xyc}\right)^\beta \left(\hat{Y}_{xyc}\right)^\alpha & \\ \log\left(1 - \hat{Y}_{xyc}\right), & \text{otherwise} \end{cases} . \tag{13}$$

In (13), α and β represent the weight parameters that are set as 2 and 4, respectively. $\hat{Y} \in [0, 1]^{(W/R)\times(H/R)\times C}$ can be seen as the prediction result; W and H are the width and height of output feature maps, respectively; R is the stride of down sampling, and C is the number of categories; x and y denote the center point coordinates in the corresponding heatmap; and c and N_c represent the certain categories and the number of center points.

Subsequently, the regression of Fourier contour parameters is supervised by L_{fd}, which minimizes the distances between two discrete contour sampling points in the spatial domain that are reconstructed from real and predicted Fourier degrees. It can be intuitive to guarantee the quality of object contour detection, which can be expressed as follows:

$$L_{fd} = \frac{1}{N_c} \sum_{n=1}^{N_c} sl_1\left(\mathrm{IFT}(T, fd), \mathrm{IFT}\left(T, \widehat{fd}\right)\right). \tag{14}$$

In (14), T denotes the number of sampled contour points; IFT represents the inverse Fourier transformation of (3); sl_1 denotes the function of SmoothL_1; and fd and $\widehat{fd}$ indicate the ground truths and predicted results of Fourier degrees.

Then, the learning of scale factor $\rho_{\max}$ is supervised by L_ρ. Similarly, it is also achieved by the SmoothL_1 function, and it can be expressed as follows:

$$L_\rho = \frac{1}{N_c} \sum_{n=1}^{N_c} sl_1\left(\rho_{\max}, \widehat{\rho_{\max}}\right). \tag{15}$$

Finally, inspired by posterior guidance [48] and in order to supervise the learning of regional guidance, the focal loss [50] is used for calculating L_{reg}, which can be expressed as follows:

$$L_{\text{reg}} = \frac{-1}{Z} \sum_{i}^{C} \begin{cases} \left(1 - \hat{M}_{xy}^i\right)^\alpha \log\left(\hat{M}_{xy}^i\right), & \text{if } M_{xy}^i = 1 \\ \left(\hat{M}_{xy}^i\right)^\alpha \log\left(1 - \hat{M}_{xy}^i\right), & \text{otherwise.} \end{cases} \tag{16}$$

In (16), $M \in [0, 1]^{W\times H}$, and Z denotes the number of positive samples when $M_{xy}^i = 1$.

IV. Experimental Analysis

In this section, three main parts, including experimental setup (Section IV-A), comparison analysis (Section IV-B), and ablation analysis (Section IV-C), are elaborated as follows.

A. Experimental Setup

1) Dataset Description: To demonstrate that the proposed UncionDet can integrate the current object detection requirements (i.e., HBB OBB and InSeg) into one challenging task of arbitrary-shape object contour detection, several optical object detection datasets with different types of annotations (i.e., DOTA [51], DIOR [52], NWPUVHR-10 (NV10) [53], [54], and iSAID [55]) and two self-built multigrained object detection datasets (i.e., MG-DOTA and MG-NV10) in the remote sensing domain are introduced to compare our proposed UncionDet with other dedicated object detectors, which would evidence that the proposed UncionDet can not only perform superior performance on currently existing object detection tasks but also exhibit a powerful ability on a new challenging multigrained object detection task. Next, the details of these datasets will be first illustrated.

a) DIOR dateset [52]: In order to verify the effectiveness of the proposed UniconDet in HBB object detection task, the version of HBB for DIOR [52] is introduced. It contains 23 463 images and 192 518 instances. These instances are annotated according to 20 common categories including AL, AT, baseball field (BF), basketball court (BC), bridge (B), chimney (C), dam (D), expressway service area (ESA), expressway toll station (ETS), golf field (GF), ground track field (GTF), harbor (HB), overpass (O), ship (S), stadium (SD), ST, tennis court (TC), train station (TS), vehicle (V), and windmill (WM). The original image size is 800×800 pixels, and the spatial resolution ranges from 0.5 to 30 m. To ensure a fair comparison with other methods, we keep the original image size and adopt the original data split of DIOR [52]. Thereby, the 11 725 remote sensing images (i.e., 50% of the dataset) are selected as the trainval set, and the remaining 11 738 images are employed for testing.

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b) DOTA dataset [51]: To prove that the proposed UniconDet is applicable to the OBB object detection task, a large-scale benchmark dataset called DOTA [51] is used for evaluation, which totally includes 2806 images. Since these images are acquired from different sensors and platforms, the sizes of images range from 800 × 800 to around 4000 × 4000 pixels and contain a variety of spatial resolutions. There are 188 282 objects annotated according to 15 common categories, containing the plane (P), baseball diamond (BD), B, GTF, small vehicle (SV), large vehicle (LV), S, TC, BC, ST, soccer ball field (SBF), roundabout (R), HB, swimming pool (SP), and helicopter (H). Particularly, there are two detection tasks in this dataset, including the tasks of HBB and OBB. In addition to the annotation format, other details remain consistent. Regarding the data division, one-half of the original images is selected as the training set, one-sixth as the validation set, besides the remaining one-third as the testing set. Notably, due to the irregular image size, the original images have to be cropped into standard image patches. Therefore, following the multiscale processing strategy of [48], [56], [57], and [58], we first resize the original images into three scales (1.3, 1.0, and 0.4) and then crop all images into 640 × 640 pixels with an overlap of 160 pixels. After being processed, there are a total of 53 919 images used for training. In the testing phase, these testing images are also resized for three scales (1.5, 1.0, and 0.5) and cropped into 1024 × 1024 pixels with an overlap of 256 pixels. Considering that the evaluation results of the testing set only can be obtained from the DOTA [51] online evaluation server, we merge the results together to restore the complete results on the original image. Finally, submitting the results to the online evaluation server of DOTA [51].

c) iSAID dataset [55]: To demonstrate the compatibility of the proposed UniconDet in the InSeg task, we additionally introduced the large-scale benchmark dataset iSAID [55], which is built based on the DOTA [51] dataset. Thereby, it shares the same 2806 images and 15 categories with the DOTA [51] dataset. Differently, it possesses 655 451 instances, which is more than those of the DOTA [51] dataset. Regarding the dataset split, it is also the same as DOTA [51]. However, in our experiments, we perform training on the train set and evaluate the performance on the validation set. Due to the irregular and large image size, we also resize the original images into two scales of (1.5, 1.0), and crop these resized images into 800 × 800 pixels with an overlap of 200 pixels. Finally, there are 24 976 images being used for training. Meanwhile, the processing of the validation set remains the same as the testing set of the DOTA [51], and the results of image patches would be combined together for evaluating the performance of the original images.

d) NV10 dataset [53], [54]: In addition to the datasets mentioned above, NV10 [53], [54] dataset with different types of annotations is also introduced for evaluating the proposed UniconDet. In detail, NV10 [53], [54] is a multiclass geospatial object detection dataset, containing a total of 800 optical aerial images, which are collected from the Google Earth server and Vaihingen data. The dataset can be divided into two sets: the positive image set, which possesses at least one instance in an image, contains 650 images, and then the negative image set, which contains 150 images without any objects. Subsequently, only the positive image set is used in the experiments. There are ten typical categories including the AL, S, ST, BD, TC, BC, GTF, HB, B, and V being labeled in each image. As for the annotation type of HBB, we continue to adopt the label information of HBB from the original NV10 dataset [53]. As for the annotation type of InSeg, Su et al. [54] have made some efforts to annotate the same objects by instance masks for ground truths. Then, in order to build the annotation type of OBB, we use the maximum bounding rectangle of the pixel-level mask for each instance as its OBB. About the data split, 70% of the positive images are used for training, and 30% are utilized for evaluating the performance. Although the size of positive images is irregular, they are all smaller than and close to 1024 × 1024 pixels. Therefore, in our experimental setting, the original images are normalized into consistent 1024 × 1024 pixels by padding the pixel values of zero on the right and down sides of each image.

e) MG-DOTA dataset: To foster the advancement of a multigrained object detection benchmark and inspire researchers to focus on unified object detector design, a self-built multigrained dataset MG-DOTA is proposed, which allocates suitable annotation types for different categories. Specifically, as mentioned above, two versions of DOTA [51] dataset and iSAID [55] all adopt the same images, and only the annotation granularity is different. Consequently, in order to construct a dataset that has access to multiple annotation granularities, we select specified annotation granularity for the instance of each category according to the characteristics of the objects. In practice, because of the annotation of iSAID [55] dataset belonging to the pixel-level mask annotation, we first adopt the iSAID development kit [55] to divide the irregular original images into 800 × 800 image patches with the overlap of 200 pixels, and transform the annotation format into several poly points, such as "classname, (x_1, y_1), (x_2, y_2), ..., (x_n, y_n)." Meanwhile, the HBB and OBB versions of DOTA [51] dataset also split the images following the same procedure as iSAID [55], while maintaining the coordinates of four vertices, such as "classname, (x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4)." After that, a total of 19 396 images are divided into the train set and 4311 images are used as the test set. Based on these images, 15 categories are split into three different annotation granularities. Among them, the SBF, H, GTF, and SP employ the format of HBB. And then, the B, SV, LV, S, TC, BC, and HB are annotated in the format of OBB. Next, the annotation of several poly points is adopted for these categories including BD, ST, R, and P. One observation is that MG-DOTA has an extremely imbalanced sample distribution among these categories; thus, it is challenging for evaluating the performance of detectors. The dataset is now available at https://github.com/ZhAnGToNG1/UniconDet.

f) MG-NV10 dataset: To further make contributions to a new benchmark of multigrained object detection, another multigrained object detection dataset MG-NV10 is self-constructed, which also contains three types of annotation

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granularities. As mentioned above, the versions of HBB, OBB, and InSeg for the NV10 dataset all share the same images and instances. The annotation format keeps the same as MG-DOTA. Therefore, it is only necessary to split the ten categories into three different granularities. Specifically, AL and BD are annotated in the format of several poly points. Then, S, TC, BC, HB, B, and V are annotated in the format of OBB. Lastly, GTF and ST are selected for using the HBB annotation. As for the data split, 455 images are chosen as the train set; another 195 images are used as the test set. The dataset has also been made available at the https://github.com/ZhAnGToNG1/UniconDet.

2) Implementation Details: In this work, the baseline detector of the proposed UniconDet is CenterNet [49]. Accordingly, the series of ResNet [78] are employed as the backbone network for comparison with other SOTA methods. Notably, ResNet-50 [78] is adopted for ablation experiments. Related to the data augmentation, the multiscale training and testing strategy is utilized for the comparison analysis to keep consistent with the other SOTA detectors. On the contrary, to evidently observe the effects of the designed modules, a plain procedure with single-scale training and testing is used for ablation analysis. Moreover, the random shift, random flip, random rotate, and color jitter are employed for model training in the comparison and ablation experiments. In view of the training schedule, all models involved in our proposed method are trained for 140 epochs. The learning rate is initially set as $1.25e^{-4}$ and reduces by half each at the 50th epoch and the 90th epoch. The Adam [79] optimizer is used in all experiments with batch size $= 10$. Finally, the experimental environment is built on PyTorch 1.7.0 and two NVIDIA RTX A6000 GPUs.

3) Evaluation Metrics: In this study, the multigrained object detection involves three different detection requirements of HBB, OBB, and InSeg. Thus, in order to ensure the unification of evaluation metrics, we follow the evaluation standard of the object detection task, where the intersection over union (IoU) between predictions and ground truths is set as 0.5 for evaluating performance. Based on the criterion, recall, precision, average precision (AP), and mean AP (mAP) are adopted, which are defined in PASCAL visual object classes criteria [80]. Among them, AP indicates the integral area under the precision and recall (P–R) curves. The remaining indexes are presented as follows:

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{17}$$

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{18}$$

$$\text{mAP} = \frac{1}{C} \sum_{i=1}^{C} \text{AP}_i \tag{19}$$

where true positive (TP) denotes the positive samples that are correctly identified, false positive (FP) represents the negative samples that are incorrectly identified as positive, false negative (FN) denotes the positive samples that are incorrectly identified as negative, and C represents the number of categories.

B. Comparison Analysis

In this section, first, the performance of single-grained object detection is evaluated on each conventional task, i.e., HBB, OBB, or InSeg, which is aimed at verifying the adaptability of our proposed UniconDet for each specific task. Moreover, to verify the multigrained object detection ability, the newly built multigrained detection datasets of MG-DOTA and MG-NV10 are used for evaluation. Meanwhile, in order to ensure a fair comparison, if those algorithms have corresponding source codes, the results can be obtained by executing the algorithms. Otherwise, the optimal performance in their published reports would be adopted for comparison. Notably, due to the lack of some certain results in reports of their detectors, the blank is replaced by "-." Furthermore, in Tables I–VIII, R-50, R-101, RS-101, R2-50, ReR-50, RX-101, CSPD-53, SOCD, and HL-104 represent the ResNet-50 [78], ResNet-101 [78], ResNeSt-101 [99], Res2Net-50 [100], ReResNet-50 [56], ResNeXt-101 [101], CSPDarknet-53 [70], SOCDetbackbone [102], and Hourglass-104 [49], respectively.

1) Results on HBB Task: Related to object detection with the HBB task, the HBB-based DIOR [52] dataset, which has 20 common categories, is chosen for evaluation. In order to verify that our proposed UniconDet can work well on HBB-based detection task, six classic object detection algorithms of the natural scene (e.g., Faster RCNN [59], Yolov5 [61], RetinaNet [50], PANet [62], Libra-RCNN [63], and DAB-DETR [60]) and nine remote sensing object detection algorithms (e.g., anchor-based RSPrompter [12], FSoD [1], MSFC-Net [2], RSADet [64], CBD-E [68], CANet [69], CoF-Net [65], EFET [66], and QETR [67]) are selected as comparisons.

As shown in the 16th∼18th rows of Table I, the proposed UniconDet has surpassed other detectors and achieved the best overall performance 77.6% mAP on the DIOR [52] dataset. Besides, when equipped with a shallower backbone network, the proposed UniconDet can still maintain competitive results in comparison. The results indicate that the unified parametric modeling strategy of FCPM has the capability to accommodate the detection task of HBB; at the same time, the designed MVCA and C^2-ERL modules can make the proposed UniconDet cope well with multiscale and multiclass object detection against complex background inferences. Thus, in Table I, by comparing the results in the first∼sixth rows with the 7th∼15th rows, one obvious tendency is that the natural scene detectors nearly perform lower mAP than the ones in the remote sensing domain. Notably, the reason why natural scene detectors often exhibit poor performance is that a naive feature extraction and fusion way in each detector is unable to adapt to the challenges of large-scale variances and complex background interferences on the DIOR [52] dataset. For instance, the Faster RCNN [59] only adopts a single-scale feature layer to classify and regress the object boundary boxes. As a result, regarding the typical co-scene categories such as AL and AT, HB and S, and ESA and V, the limited receptive field of the single-scale feature layer cannot capture suitable representations for these objects with significant scale difference, so as to obtain the lowest mAP on

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TABLE I
COMPARISON EXPERIMENTS ABOUT *HBB* TASK ON DIOR [52]

	Method	Backbone	mAP	AL	AT	BF	BC	B	C	D	ESA	ETS	GF	GTF	HB	O	S	SD	ST	TC	TS	V	WM
Natural Scene	Faster RCNN [59]	VGG-16	54.1	53.6	49.3	78.8	66.2	28.0	70.9	62.3	69.0	55.2	68.0	56.9	50.2	50.1	27.7	73.0	39.8	75.2	38.6	23.6	45.4
	DAB-DETR [60]	R-50	66.7	76.1	80.9	78.0	87.2	40.3	80.2	65.3	73.9	57.3	78.7	78.9	38.6	56.5	55.3	65.5	47.4	87.1	64.4	44.4	86.0
	Yolov5 [61]	DarkNet-53	68.5	87.2	61.6	73.7	90.0	42.6	77.5	55.2	63.8	63.2	66.8	78.0	58.2	58.1	87.8	54.3	79.3	89.7	50.2	53.9	79.6
	RetinaNet [50]	R-101	65.2	63.4	47.8	83.3	78.3	48.2	67.9	49.4	53.1	73.6	66.2	59.1	47.5	45.6	72.0	82.3	80.7	90.1	54.9	47.7	**92.9**
	PANet [62]	R-101	66.1	60.2	72.0	70.6	80.5	43.6	72.3	61.4	72.1	66.7	72.0	73.4	45.3	56.9	71.7	70.4	62.0	80.9	57.0	47.2	84.5
	Libra RCNN [63]	R-101	70.7	60.8	79.8	71.7	87.8	49.5	79.8	64.8	82.0	74.8	79.7	82.5	42.9	63.0	72.0	75.4	62.7	81.3	64.3	49.9	88.5
Remote Sensing	RSADet [64]	DLA-34	72.2	73.6	86.0	72.6	89.6	43.6	75.3	62.3	79.5	68.7	78.6	79.1	57.9	59.2	**90.0**	55.8	77.0	87.8	65.3	55.3	86.5
	RSPrompter [12]	R-50	58.5	77.2	50.2	75.5	72.8	25.6	66.8	37.8	59.1	53.1	62.8	63.3	41.4	50.9	70.2	64.6	53.1	80.0	52.8	35.8	76.6
	CoF-Net [65]	R-50	75.8	84.0	85.3	82.6	90.0	47.1	80.7	73.3	**89.3**	74.0	84.5	83.2	57.4	62.2	82.9	77.6	68.2	89.9	68.7	49.3	85.2
	EFET [66]	R-50	71.3	68.6	72.3	70.9	82.1	**80.6**	76.6	62.2	71.8	73.3	75.6	70.2	**65.2**	50.1	81.3	74.3	73.3	81.0	63.3	55.1	78.6
	QETR [67]	R-50	71.5	73.3	**90.3**	77.6	88.2	47.2	**82.5**	**76.6**	86.9	72.8	**86.4**	84.4	51.0	62.2	51.8	**84.2**	39.7	85.0	**71.8**	39.3	85.9
	FSoD [1]	R-101	71.8	88.9	66.9	**86.8**	90.2	45.5	79.6	48.2	86.9	75.5	67.0	77.3	53.6	59.7	78.3	69.9	75.0	**91.4**	52.3	52.0	90.6
	CBD-E [68]	R-101	67.8	54.2	77.0	71.5	87.1	44.6	75.4	63.5	76.2	65.3	79.3	79.5	47.5	59.3	69.1	69.7	64.3	84.5	59.4	44.7	83.1
	CANet [69]	R-101	74.3	70.3	82.4	72.0	87.8	55.7	79.9	67.7	83.5	77.2	77.3	83.6	56.0	**63.6**	81.0	79.8	70.9	88.2	67.6	51.2	89.6
	MSFC-Net [2]	RS-101	70.1	85.8	76.2	74.3	90.1	44.1	78.1	55.5	60.9	59.5	76.9	73.6	49.5	57.2	89.6	69.1	76.5	86.7	51.8	55.2	84.3
	UniconDet (Ours)	R-50	75.8	87.9	78.2	85.3	**90.8**	49.8	80.0	59.7	80.9	78.6	79.0	82.9	57.8	62.3	88.2	81.9	78.3	89.6	59.0	59.3	85.2
		R-101	76.6	87.1	82.9	86.4	90.5	50.8	78.4	65.2	81.4	78.5	80.5	**84.4**	59.3	60.2	88.2	80.2	80.0	89.8	61.9	59.6	85.2
		R-152	**77.6**	**89.7**	85.5	86.7	90.5	52.9	79.3	69.2	82.6	**80.2**	80.9	83.8	59.1	63.2	87.5	76.8	**80.8**	90.1	65.2	**60.8**	87.0

the DIOR [52] dataset. Although RetinaNet [50], PANet [62], Yolov5 [61], and Libra RCNN [63] design the multiscale regression layers for solving this issue, the plain feature fusion way still fails to resist the complex background interferences and then restrict the upper bound performance for HBB object detection in remote sensing scenes. In contrast, CoF-Net [65] proposes a coarse-to-fine manner for multispectral nonlocal feature adaptation, which distinctively enriches spatial feature details. Meanwhile, CoF-Net [65] also considers to chose the high-quality sample allocation for remote sensing objects with different scale distributions. As a result, CoF-Net achieves an excellent performance of 75.8% mAP, as shown in the ninth row of Table I. Apart from the conventional detection architecture, the transformer-based SOTA detectors (i.e., DAB-DETR [60] and QETR [67]) have also been introduced for comparison. As shown in the second row of Table I, DAB-DETR achieves the unsatisfactory 66.7% mAP on the DIOR [52] dataset. The results indicate that transformer-based detectors consider the global interaction of local elements, but have a poor ability to perceive local details about small-scale objects. This point can be proved according to the AP of S and V. For instance, as shown in the 11th row of Table I, although QETR [67] exceeds DAB-DETR [60] 4.8% mAP from the overall performance, the reported APs about S and V both obtain very bad performance, which shows that transformer-based detectors still have huge room to improve performance for small-scale object detection. Furthermore, on the whole, the advanced results of DIOR [52] dataset are still dominated by these CNN-based detectors, such as RSADet [64], CANet [69], and CoF-Net [65]. Indeed, the global modeling capability of the transformer is undeniably valuable, offering a complementary effect to the local receptive field of CNNs. To this end, in our proposed method, the cross-attention mechanism of the transformer network is introduced into the framework of a CNN-based detector to design the MVCA module, aiming to advance the perception of Fourier contour parameters.

In addition, the NV10 [53] dataset is also considered to be used for the performance evaluation of our proposed UniconDet under the task of HBB detection. Then, the comparison results are reported in Table II. We can see that our proposed UniconDet can achieve the best overall performance of 95.60% mAP, while obtaining the highest 98.36% AP on the class of V. This demonstrates that by adopting the unified parametric modeling way of FCPM, the designed MVCA and C^2-ERL can make the proposed UniconDet tackle small object detection under the task of HBB detection. As for other comparisons, similar to the above findings according to the DIOR [52] dataset, Faster-RCNN [59] achieves the worst overall performance of 84.50% mAP. Although it can perform good detection performance for large-scale objects, i.e., the 100.0% AP for GTF, the inferior results are presented for small-scale object detection, such as S and V shown in Table II. As shown in the 6th∼13th rows of Table II, the designed detectors in the remote sensing domain are meticulously considered for the characteristics of remote sensing objects; thus, they can obtain a large increase in overall mAP by the newly designed attention mechanisms, feature fusion modules, and so on. However, there is still considerable room for improvement in small-scale object detection such as the category V. Actually, in addition to improving the detection performance of small objects from the perspective of feature extraction, the reasonable parametric modeling strategy also plays a critical role. The reason is that under the conventional modeling way of (x, y, w, h), a slight prediction error of a certain parameter value would cause a significant shift from the ground truth of small-scale objects. Instead, our designed parametric modeling strategy of FCPM not only can eliminate the scale difference by object scale normalization, but also scatters the prediction errors of discrete contour sampling points into compact Fourier contour parameters. This can avoid the collapse of prediction errors for the conventional modeling way of (x, y, w, h). At the same time, as the results reported in Table II, the designed effective feature extraction way of MVCA and the regression layer of C^2-ERL can be effectively combined to support the Fourier contour parameter learning of HBB according to FCPM.

2) Results on OBB Task: Under the remote sensing scene, since some objects within specific categories have significant directional information, the OBB detection task has emerged a hot research topic for achieving more accurate

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TABLE II
COMPARISON EXPERIMENTS ABOUT *HBB* TASK ON NV10 [53]

	Method	Backbone	mAP	AL	S	ST	BD	TC	BC	GTF	HB	B	V
Natural Scene	Faster RCNN [59]	R-50	84.50	90.90	86.30	90.50	98.20	89.70	69.60	100.0	80.10	61.50	78.10
	DETR [43]	R-50	92.90	93.00	80.50	95.50	99.40	90.20	**99.30**	99.00	99.10	81.10	92.20
	YOLOv8 [70]	CSPD-53	93.61	99.89	94.01	96.67	97.93	93.90	90.82	96.79	90.65	84.69	90.60
	RetinaNet [50]	R-101	90.90	98.60	75.20	89.70	99.10	87.90	99.10	97.20	98.30	86.70	77.40
	FoveaBox [71]	R-101	91.42	99.48	83.60	96.83	95.14	85.65	84.37	95.54	90.78	81.67	91.15
Remote Sensing	GLASANet [72]	R-50	94.57	99.90	**95.80**	97.12	**99.40**	**98.75**	86.12	99.46	97.51	84.83	86.76
	RSPrompter [12]	R-50	92.90	99.80	89.50	98.60	98.60	91.10	85.00	99.90	86.40	88.20	92.30
	EFET [66]	R-50	94.60	97.20	83.70	96.20	96.70	95.10	97.80	96.20	93.20	96.20	93.20
	MSGN [73]	CSPD-53	95.53	98.89	93.68	97.90	98.60	91.82	92.02	99.98	99.09	91.07	92.20
	Yolov3-Att [74]	Darknet-53	94.49	100.0	91.60	**99.93**	94.70	91.97	86.62	97.52	94.80	93.18	94.59
	CAD-Net [75]	R-101	91.50	97.00	77.90	95.60	93.60	87.60	87.10	99.60	**100.0**	86.20	89.90
	DA^{2}FNet [76]	HL-104	94.58	99.92	89.48	87.18	98.65	95.66	95.24	99.48	98.96	88.98	93.56
	MashFormer [77]	HL-104	93.80	**100.0**	93.43	92.64	96.91	94.56	95.56	100.0	79.52	94.08	91.30
	UniconDet (Ours)	R-50	95.26	99.95	91.44	98.41	98.31	83.16	93.97	100.0	94.04	**100.0**	93.36
		R-101	**95.60**	99.99	93.70	97.20	95.01	92.53	93.81	99.88	97.86	91.43	94.67
		R-152	95.29	99.99	93.82	98.72	98.62	82.17	94.05	**100.0**	91.58	95.56	**98.36**

TABLE III
COMPARISON EXPERIMENTS ABOUT *OBB* TASK ON DOTA [51]

	Method	Backbone	mAP	P	BD	B	GTF	SV	LV	S	TC	BC	ST	SBF	R	HB	SP	H
Two-stage	APE [81]	R-50	75.75	89.96	83.62	53.42	76.03	74.01	77.16	79.45	90.83	87.15	84.51	67.72	60.33	74.61	71.84	65.55
	ReDet [56]	ReR-50	80.10	88.81	82.48	60.83	80.82	78.34	86.06	88.31	90.87	88.71	87.03	68.65	66.90	79.26	79.71	76.67
	ORSR [58]	ReR-50	80.78	89.77	84.22	61.49	**82.51**	78.76	**87.11**	**89.45**	**91.86**	88.67	86.77	**72.68**	67.58	79.43	78.74	72.68
	RoI Trans. [82]	R-101	69.56	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67
	FR-O [51]	R-101	54.13	79.42	77.13	17.70	64.05	35.30	38.02	37.16	89.41	69.64	59.28	50.30	52.91	47.89	47.40	46.30
	Gliding Ver. [83]	R-101	75.02	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	70.91	72.94	70.86	57.32
	MaskOBB [84]	RX-101	75.33	89.56	85.95	54.21	72.90	76.52	74.16	85.63	89.85	83.81	86.48	54.89	69.64	73.94	69.06	63.32
	AOPG [57]	R-101	80.19	89.99	86.14	60.20	79.55	78.47	84.93	88.79	90.88	87.32	87.07	71.50	71.22	83.57	72.47	70.77
	AQE-Det [85]	R-101	78.35	89.49	85.89	55.46	77.32	74.17	80.33	87.65	90.82	86.99	86.52	66.58	69.06	77.12	78.35	69.51
	STD-O [86]	ViT-B	**81.66**	88.56	84.53	62.08	81.80	81.06	85.06	88.43	90.59	86.84	86.95	72.13	71.54	**84.30**	82.05	78.94
One-stage	AF2Det [87]	R2-50	78.30	89.71	84.08	53.84	74.76	79.05	85.18	88.43	90.82	88.71	87.10	67.74	67.30	76.86	71.93	71.98
	AOGC [88]	R-50	76.55	83.44	80.29	54.06	70.90	**81.52**	83.42	88.24	90.88	83.02	86.84	60.34	64.90	75.36	80.23	64.80
	TIOE-Det [89]	R-50	78.69	89.76	85.23	56.32	76.17	80.17	85.58	88.41	90.81	85.93	87.27	68.32	70.32	68.93	78.33	68.87
	GGHL [90]	DarkNet-53	76.95	89.74	85.63	44.50	77.48	76.72	80.45	86.16	90.83	88.18	86.25	67.07	69.40	73.38	68.48	70.14
	DRN [91]	HL-104	73.23	89.71	82.34	47.22	64.10	76.22	74.43	85.84	90.57	86.18	84.89	57.65	61.93	69.30	69.63	58.48
	SCRDet++ [92]	R-101	79.35	90.00	**86.25**	**65.04**	74.52	72.93	84.17	79.05	90.72	87.37	87.06	72.10	66.72	82.64	80.57	71.07
	IENet [93]	R-101	57.14	80.20	64.54	39.80	32.07	49.71	65.01	52.58	81.45	44.66	78.51	46.54	56.73	64.40	64.24	36.75
	BBAVector [4]	R-101	75.36	88.63	84.06	52.13	69.56	78.26	80.40	88.06	90.87	87.23	86.39	56.11	65.62	67.10	72.08	63.96
	Axis learning. [94]	R-101	65.98	79.53	77.15	38.59	61.15	67.53	70.49	76.30	89.66	79.07	83.53	47.27	61.01	56.28	66.06	36.05
	PIIDet [48]	RS-101	80.21	89.67	86.04	55.76	70.64	81.35	85.85	88.40	90.79	88.37	**88.16**	66.91	72.40	77.74	81.95	**79.05**
	CSL [3]	R-152	76.17	**90.25**	85.53	54.64	75.31	70.44	73.51	77.62	90.84	86.15	86.69	69.60	68.04	73.83	71.10	68.93
	GCL [95]	Swin-T	81.41	89.40	85.19	60.39	79.88	81.37	85.80	88.99	90.85	85.73	88.02	70.38	**73.52**	83.56	**82.49**	75.63
	UniconDet (Ours)	R-50	78.40	88.33	81.86	52.12	70.57	77.22	84.09	88.46	90.77	87.78	88.00	66.04	69.11	75.55	79.98	76.13
		R-101	78.45	88.28	84.35	52.02	72.86	79.19	85.14	88.18	90.83	**88.72**	87.79	69.97	67.21	76.82	71.08	74.33
		R-152	78.67	89.00	82.22	51.31	70.42	77.55	83.93	88.46	90.80	87.85	88.03	68.41	68.98	75.81	79.79	77.55

object detection. Different from the HBB detection task, apart from the vast scale variance and complex background interference, the arbitrary orientation prediction has become a new challenge. Therefore, in order to explore the adaptability of our proposed UniconDet on the OBB detection task, a large-scale multiclass object detection dataset called DOTA [51] is introduced for evaluation. Meanwhile, there are 22 detectors specifically designed for OBB detection that have been chosen for comparison, which includes one-stage and two-stage paradigms. Then, the results are reported in the 23rd∼25th rows of Table III. Notably, under the more concise one-stage anchor-free object detection architecture, the proposed UniconDet obtains approachable SOTA results (e.g., 78.40%, 78.45%, and 78.67% mAP) based on the backbone networks of R-50, R-101, and R-152. The results demonstrate that, except for being able to cope with vast scale variances, the proposed UniconDet can effectively avoid the problem of discontinuous angle regression for the OBB detection task, without adding additional processes and increasing model complexity. Then, from the fourth and fifth rows of Table III, RoI Trans [82] and FR-O [51] are designed for the OBB detection task by directly adding the prediction of angle or corner points into the HBB detector (i.e., Faster RCNN [59]); however, they present poor overall performance, i.e., 69.56% and 54.13% mAP. The reason for this phenomenon is due to unreliable parametric modeling and invalid

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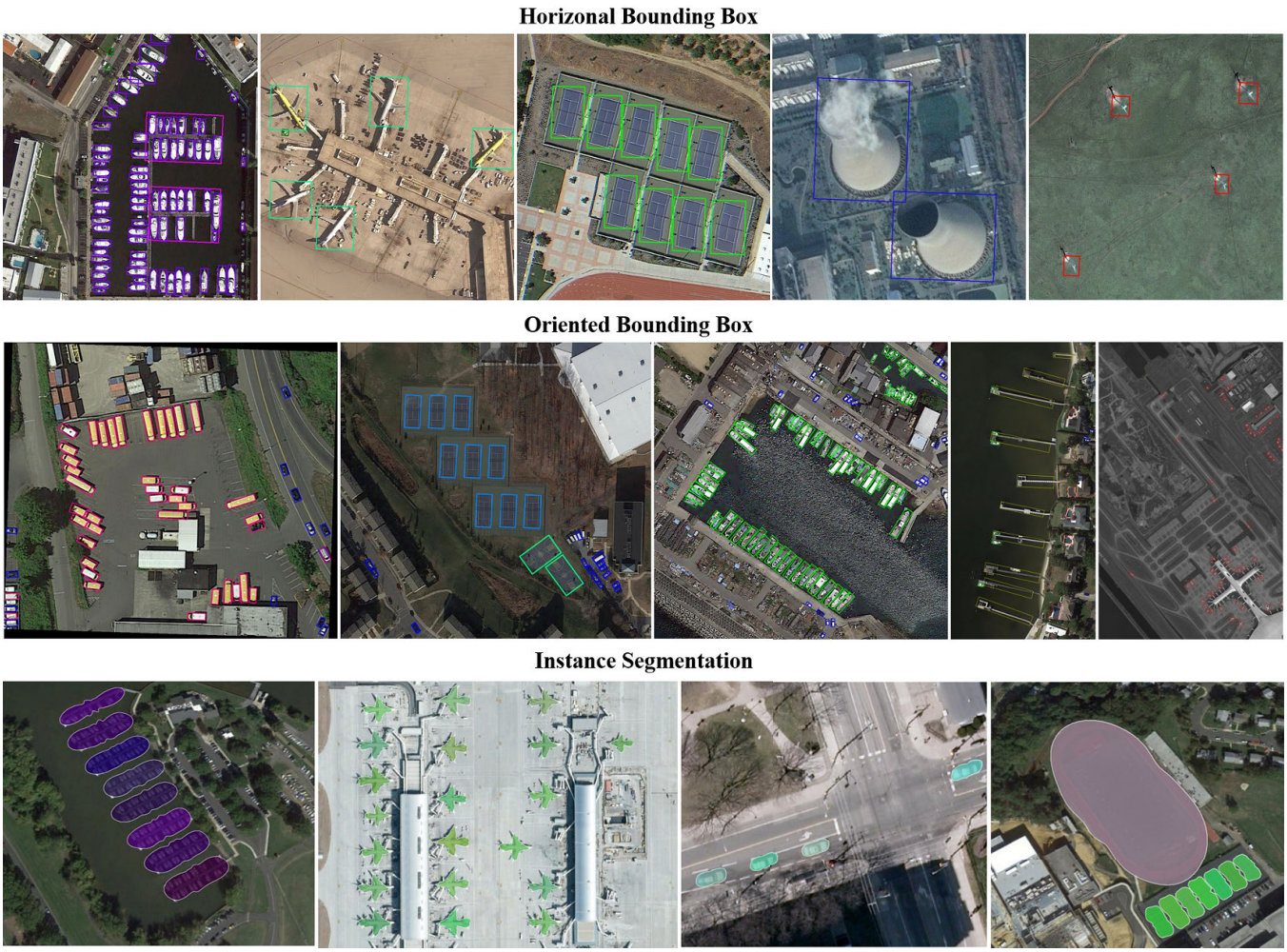


Fig. 6. Detection results of the proposed UniconDet on different benchmarks (e.g., HBB, OBB, and InSeg).

feature perception for the rotated objects. To tackle this issue, some research has been devoted to redesigning more reasonable OBB parametric definitions, such as MaskOBB [84], APE [81], Gliding Ver [83], IENet [93], BBAvector [4], and Axis learning [94], which can achieve significant performance improvement. Specifically, these methods would decompose a single angle parameter into several customized parameters to avoid original discontinuous angle periodic regression. Hence, it inevitably brings extra computations and postprocessing in the design of the detectors. Moreover, regarding the perception of rotated objects, many SOTA detectors, such as ReDet [56], ORSR [58], and AOPG [57], adopt a two-stage paradigm and generate high-quality rotated proposals by specified feature alignment design. Therefore, as reported in the second, third, and eighth rows of Table III, the OBB detectors of ReDet [56], ORSR [58], and AOPG [57] achieve the SOTA overall performance, i.e., 80.10%, 80.78%, and 80.19% mAP. In addition, from the 10th and 22nd rows of Table III, we can find that STD-O [86] and GCL [95] both adopt an advanced transformer backbone with pretraining technology for detectors, which can perform excellent results, i.e., 81.66% and 81.41% mAP because of the powerful basic feature extraction ability. Although the abovementioned detectors can achieve higher OBB detection performance, extra heavy computation and memory consumption would be needed. As a comparison, the proposed UniconDet is not sensitive to backbone changes, which is because of the powerful parametric modeling strategy of FCPM, the specific feature extraction way of MVCA, and the regression layer of C^2-ERL. Meanwhile, as all of results reported in Table III, one observation is that the proposed UniconDet can adapt well to OBB detection task, and provide toward SOTA performance without multistage refinement and powerful foundation models.

Additionally, to verify the robustness of the proposed UniconDet, the NV10 [53] dataset annotated with the granularity of OBB is also introduced for evaluation. Meanwhile, there are nine classic OBB detectors selected for comparison. As reported in the 10th∼12th rows of Table IV, it can be found the proposed UniconDet can achieve the highest overall performance of 94.41% mAP compared with other methods. Under different backbone network equipment, the proposed UniconDet also presents a stable overall performance. Notably, when coping with the multiclass object detection task, several categories are often exhibited with the characteristic of extremely high aspect ratios such as S, HB, and B, which are very sensitive to the changes of IoU in the performance evaluation.

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TABLE IV
COMPARISON EXPERIMENTS ABOUT *OBB* TASK ON NV10 [54]

	Method	Backbone	mAP	AL	S	ST	BD	TC	BC	GTF	HB	B	V
Two-stage	ReDet [56]	ReR-50	91.40	90.70	90.20	**99.60**	90.80	89.40	**98.20**	99.80	77.70	87.30	90.40
	Gliding Ver. [83]	R-101	72.32	89.90	15.20	90.80	90.50	66.90	80.90	88.80	60.60	46.10	88.10
	ORPN [96]	R-101	81.80	99.60	43.60	90.90	90.50	89.20	90.60	97.70	81.50	45.20	88.90
	STD-O [86]	ViT-B	85.35	90.50	78.40	90.80	90.80	87.30	77.90	99.20	77.50	73.40	87.70
One-stage	GCL [95]	R-50	85.45	99.49	69.89	90.88	90.75	90.60	80.91	98.93	77.85	65.82	89.38
	TIOE-Det [89]	R-50	70.27	96.00	50.74	98.45	90.44	67.45	28.78	87.74	75.77	31.19	76.16
	S2ANet [97]	R-101	86.90	99.60	68.80	90.70	90.70	87.60	80.10	99.60	78.90	82.90	90.10
	R3Det [98]	R-101	81.63	98.60	70.10	90.70	90.30	79.40	81.20	99.00	64.60	53.30	89.20
	CSL [3]	R-101	76.92	**99.70**	15.80	90.80	90.50	78.30	81.20	99.60	80.90	51.50	80.70
	UniconDet (Ours)	R-50	94.07	99.63	90.65	97.90	96.25	**94.65**	96.78	99.10	88.62	80.85	96.28
		R-101	94.40	98.61	91.70	98.49	**96.62**	92.53	97.03	**99.84**	85.83	**89.16**	94.41
		R-152	**94.41**	99.63	**93.73**	98.85	96.04	91.95	94.05	99.61	**89.42**	84.08	**96.40**

Hence, the results of these categories convincingly reflect the quality of detected OBBs. As reported in the 5th, 11th, and 12th columns of Table IV, the proposed UniconDet obtains the highest single-class AP on the categories of S, HB, and B. The reason is that our proposed UniconDet achieves the estimation of orientations by geometrical contour modeling rather than conventional angle regression strategies, which are employed by other methods (e.g., CSL [3], ORPN [96], R3Det [98], and ReDet [56]). Thus, due to more unified parametric modeling, robust feature extraction, and stable regression, it makes the proposed UniconDet not only effectively avoid the problem of discontinuous angle periodic regression but also provide more robust OBB detection performance for objects with extremely high aspect ratios. Moreover, as shown in the fourth, fifth, and sixth rows of Table IV, the SOTA detectors of STD-O [86], GCL [95], and TIOE-Det [89] in DOTA [51] obtained unremarkable results in NV10 [53], which indicates the instability of OBB parameter regression for objects with extremely high aspect ratios. Besides, ReDet [56] still maintains a more competitive overall performance, i.e., 91.40% mAP in the first row of Table IV. We analyze that the key reason is the multiple customized solutions for the OBB detection task, including dense rotated anchor design, a rotation-equivariant backbone network, and a rotation-invariant RoI align module, are designed for ReDet [56]. As a result, these designed modules combined together can ensure the accuracy maintenance both in DOTA [51] and NV10 [53]. Even so, whether in terms of inference efficiency or performance, our proposed UniconDet still holds superiority in the OBB detection task of the NV10 [53] dataset.

3) Results on InSeg Task: In this section, we further exploit the performance of our proposed UniconDet in the InSeg task, which involves two remote sensing datasets, i.e., iSAID [55] and NV10 [54]. According to these two datasets, a variety of pixel-based and contour-based InSeg algorithms are introduced as comparisons, and then, they are all designed for the natural scene (e.g., MaskRCNN [103], Mask2Former [15], Mask-Scoring (MS) RCNN [104], PANet [62], QueryInst [105], E2EC [32], PolySnake [36], and YOLACT [109]) and the remote sensing scene (e.g., R-ARE-Net [106], DCTC [108], SS-PANet+ [107], SS-MASK-RCNN [107], RSPrompter [12], and CATNet [9]).

First, as mentioned above, the iSAID [55] dataset adopts the same images as the DOTA [51] dataset; however, an extra 457 169 small-scale instances are further annotated, involving the categories of SV, LV, S, and ST. These challenging dataset-specific categories will be discussed in detail as follows. Then, as reported in the 13th∼15th rows of Table V, the proposed UniconDet can achieve the best overall performance of 66.30% mAP. At the same time, regarding more challenging dataset-specific categories, the proposed UniconDet also achieved the highest AP for S and ST, i.e., 83.89% AP on S and 72.17% AP on ST, and the second highest AP for SV and LV, i.e., 43.03% AP on SV and 66.41% AP on LV. These results indicate that the designed unified parametric modeling strategy of FCPM can make the proposed UniconDet have the capability of fine-grained InSeg. However, as the results shown in the 16th and 18th columns of Table V, there seem to be some defects that exist in the proposed UniconDet for certain categories, i.e., H and HB, then resulting in the lower AP in the InSeg task. The reason is that the current discrete contour sampling mechanism makes the designed FCPM not very suitable for the instances that have nonconvex contours. Differently, the pixel-based methods only attend to the pixel-level prediction of each class so that they can describe the objects with extreme nonconvex contours. However, if only focusing on pixel-level prediction, it would lead to a lack of understanding of the overall objects, presenting fragmentary pixel-level classification results. Thus, as reported in Table V, these pixel-based methods, i.e., MaskRCNN [103], RSPrompter [12], CATNet [9], Mask2Former [15], MS RCNN [104], PANet [62], QueryInst [105], R-ARE-Net [106], and SS-PANet+ [107] often present limited overall performance for the multiclass InSeg task.

In order to further analyze the performance of pixel-based and contour-based methods, from Table V, we can find that the pixel-based methods such as MaskRCNN [103], MS RCNN [104], and PANet [62] from natural scenes would obtain more similar overall performance with contour-based methods, i.e., E2EC [32] and PolySnake [36]. However, when facing more challenging InSeg detection for small-scale

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TABLE V
COMPARISON EXPERIMENTS ABOUT *InSeg* TASK ON iSAID [55]

	Method	Backbone	mAP	P	BD	B	GTF	SV	LV	S	TC	BC	ST	SBF	R	HB	SP	H
Pixel-based	MaskRCNN [103]	R-50	62.10	85.80	83.30	53.30	55.30	23.60	59.70	63.60	90.90	62.60	55.40	**69.30**	65.30	67.50	66.40	30.00
	RSPrompter [12]	R-50	63.50	81.60	83.20	50.30	58.90	**51.00**	**79.80**	73.00	88.60	46.50	58.30	40.40	59.20	**75.90**	58.00	**48.00**
	CATNet [9]	R-50	63.60	87.30	80.80	52.70	54.80	30.20	63.60	80.40	91.00	60.60	59.20	66.50	66.10	69.00	64.40	26.90
	Mask2Former [15]	R-50	54.60	76.40	76.80	51.20	53.00	20.20	49.80	54.00	87.30	55.70	52.00	62.50	49.80	61.80	55.70	12.90
	MS RCNN [104]	R-101	62.00	84.70	83.10	50.40	57.90	24.40	61.50	64.50	91.80	62.70	57.00	69.00	**67.00**	67.50	66.10	22.00
	PANet [62]	R-101	60.50	86.40	82.40	51.60	53.90	23.60	60.00	63.60	91.60	58.60	57.30	66.10	64.40	67.50	62.40	18.10
	QueryInst [105]	R-101	56.90	80.20	77.90	49.40	55.40	23.30	50.80	56.30	87.90	55.10	56.30	53.80	54.00	61.90	60.10	30.20
	R-ARE-Net [106]	R-101	60.80	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	SS-PANet+ [107]	R-101	65.60	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Contour-based	DCTC [108]	DLASeg	58.20	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	E2EC [32]	DLA-34	60.20	85.30	80.16	49.65	42.91	29.81	60.43	64.55	90.54	58.05	62.73	51.89	63.57	64.97	70.61	28.17
	PolySnake [36]	DLA-34	61.70	87.25	80.74	53.22	52.07	29.27	59.78	64.06	90.50	58.24	63.26	58.62	65.31	65.19	72.72	25.77
	UniconDet (Ours)	R-50	65.27	89.97	84.85	50.98	67.46	43.03	66.22	83.85	93.87	73.07	**72.17**	67.05	63.08	29.53	73.60	20.43
		R-101	**66.30**	90.10	**88.52**	**53.57**	66.89	43.01	66.41	83.74	**94.64**	**73.67**	70.17	65.96	66.83	30.95	74.88	25.15
		R-152	66.20	**91.01**	86.80	52.02	**71.57**	42.57	66.12	**83.89**	93.79	75.10	70.91	66.56	65.05	33.73	**75.49**	18.21

objects (i.e., SV and ST), there is a significant performance gap between pixel-based and contour-based methods. As shown in the 11th and 12th rows of Table V, E2EC [32] and PolySnake [36] achieve 29.81% AP and 29.27% AP on the SV, while obtaining 62.73% AP and 63.26% AP on the ST, and their results would surpass the pixel-based methods (e.g., MaskRCNN [103], MS RCNN [104], PANet [62], and QueryInst [105]) by about 5% and 6% AP on the SV and ST, respectively. Consequently, according to small-scale object detection, the contour-based methods can provide more advantageous InSeg detection performance than pixel-based methods. Furthermore, due to the development of semantic segmentation technology, there are some studies adapting a two-stage detection paradigm or designing specific modules to improve the InSeg detection performance of pixel-based methods. For example, SS-PANet+ [107] is built upon the PANet [62] and introduces an additional scale complementary mask branch to improve the InSeg detection performance, which can achieve 65.6% mAP, as shown in Table V. CATNet [9] focuses on the perception of object scale, and therefore, a multiscale context aggregation module is designed in CATNet [9] to improve multiscale perceptive ability, then facilitating the InSeg task, which can get a tremendous performance gain on the category of S, exceeding other chosen comparisons by about 16%∼26% AP. Moreover, referring to the remarkable semantic segmentation ability of the SAM [113], RSPrompter [12] leverages the foundation SAM model to be incorporated with semantic class information for achieving the InSeg detection from remote sensing imagery (RSI), which can achieve acceptable performance on the iSAID [55] dataset. However, although there are more great progresses in pixel-based methods, they require more annotation labor and computational complexity, which restricts inference efficiency and further accuracy improvement for the InSeg task. As the results shown in Table V, our proposed UniconDet can provide superior and stable overall performance (e.g., 65.27%, 66.30%, and 66.20% mAP) without being influenced by the changes of various backbones, which means that the proposed UniconDet can cope well with the InSeg detection requirement.

To demonstrate the robustness of the proposed UniconDet on the InSeg task, the pixel-level annotated NV10 [54] dataset is further employed for comparison. As reported in Table VI, our proposed UniconDet still achieves the highest performance of 94.90% mAP. Moreover, the remote sensing InSeg detectors (e.g., R-ARE-Net [106], CATNet [9], and RSPrompter [12]) also exhibit outstanding performance on the NV10 [54] dataset of the InSeg task. Furthermore, other InSeg detectors from natural scenes also achieve good results. This indicates that compared with the iSAID [55] dataset, there are many challenging small-scale instances that are not annotated. Under these data conditions, although pixel-based and contour-based methods both perform well on the NV10 [54] dataset of InSeg task, the pixel-based methods would easily lead to fragmentary pixel-level classification so that they present a big risk of providing unsteady InSeg results. For example, the famous Mask2Former [15], which can unify different semantic segmentation tasks in natural scene, performs the worst overall performance on the iSAID [55] dataset but achieves the approachable SOTA result on the NV10 [54] dataset. This phenomenon means that some pixel-based methods like Mask2Former [15] would lack stability for datasets with different data characteristics.

At the end, in order to observe the detection performance of our proposed UniconDet in three detection tasks (e.g., HBB, OBB, and InSeg) more intuitively, the visualization results have been shown in Fig. 6. By directly observing these visualized results, one conclusion can be drawn: the proposed UniconDet can answer the various granularity detection requirements (i.e., HBB, OBB, and InSeg), and it can be a promising and feasible unified object detector in the remote sensing domain.

4) Results on Multigrained Task: As mentioned above, different detection requirements (i.e., HBB, OBB, and InSeg) should be considered for each class of objects in achieving remote sensing object detection at different granularities, which allows the detector to focus on detecting the specific objects in a very simple yet suitable way. It is beneficial for practical applications. Therefore, to assess the capacity of the proposed UniconDet on the multigrained object detection benchmark, two self-built

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TABLE VI
COMPARISON EXPERIMENTS ABOUT *InSeg* TASK ON NV10 [54]

	Method	Backbone	mAP	AL	S	ST	BD	TC	BC	GTF	HB	B	V
Pixel-based	MaskRCNN [103]	R-50	91.00	92.00	84.50	98.80	98.70	84.10	75.50	99.50	89.90	94.00	92.80
	Mask2Former [15]	R-50	92.40	98.20	85.40	94.30	97.70	84.30	87.40	100.0	86.00	95.90	91.00
	QueryInst [105]	R-50	72.80	83.80	75.10	53.80	87.20	56.00	62.70	98.20	67.70	72.70	71.20
	PANet [62]	R-50	89.50	91.90	82.50	98.30	98.70	83.40	78.30	99.50	84.00	86.10	92.90
	CATNet [9]	R-50	93.70	**98.80**	89.60	**99.80**	98.20	84.60	95.50	99.90	89.50	86.00	94.90
	RSPrompter [12]	R-50	93.30	97.60	89.20	97.80	98.30	90.20	80.10	99.30	91.40	**98.90**	89.70
	MS RCNN [104]	R-101	90.50	93.20	85.80	98.00	98.60	87.40	79.50	98.00	88.60	85.90	90.10
	YOLACT [109]	R-101	83.40	56.30	77.10	97.60	97.80	54.40	82.90	98.00	84.90	96.10	89.50
	SS-Mask-RCNN [107]	R-101	92.00	-	-	-	-	-	-	-	-	-	-
	R-ARE-Net [106]	R-101	93.20	-	-	-	-	-	-	-	-	-	-
Contour-based	DCTC [108]	DLASeg	91.80	-	-	-	-	-	-	-	-	-	-
	E2EC [32]	DLA-34	93.60	95.96	88.19	96.95	**99.27**	91.40	93.84	97.91	94.59	86.62	90.87
	PolySnake [36]	DLA-34	90.40	96.13	91.37	94.48	97.08	88.97	86.31	93.59	82.52	91.13	82.63
	DANCE [110]	R-101	90.20	87.79	87.40	98.30	98.15	86.05	81.70	95.17	89.89	89.99	87.69
	UniconDet (Ours)	R-50	**94.90**	94.36	93.82	98.72	97.34	92.78	91.08	99.84	**94.88**	89.32	**96.96**
		R-101	94.46	89.92	95.20	98.83	98.43	**93.42**	**95.61**	99.68	91.43	86.48	95.58
		R-152	94.63	92.76	**95.23**	98.82	98.69	93.04	94.05	**100.0**	88.71	89.32	95.96

TABLE VII
COMPARISON EXPERIMENTS ABOUT MULTIGRAINED DETECTION TASK ON THE MG-DOTA. * RED: INSEG; GREEN: OBB; AND YELLOW: HBB

	Method	Backbone	mAP	P	BD	ST	R	B	SV	LV	S	TC	BC	HB	GTF	SBF	SP	H
Pixel-based	PANet [62]	R-50	71.00	86.90	86.60	59.30	67.10	55.00	60.50	81.30	74.70	**94.50**	73.20	**78.60**	74.20	64.50	59.20	49.40
	MaskRCNN [103]	R-50	69.30	85.00	82.10	58.50	64.40	52.90	57.70	80.50	73.80	93.90	68.40	75.20	75.40	64.00	54.00	53.80
	Mask2Former [15]	R-50	65.40	84.20	85.60	58.90	67.40	60.90	46.00	75.00	62.10	92.30	57.10	70.40	69.80	53.80	59.30	38.10
	QueryInst [105]	R-50	68.00	84.80	83.60	59.50	66.30	59.70	49.70	74.40	62.80	94.10	64.20	75.50	71.60	56.40	61.60	55.60
	CATNet [9]	R-50	73.00	87.50	84.80	59.50	72.20	71.60	84.70	**93.40**	91.90	70.90	59.50	64.00	65.20	80.50	63.40	45.90
	RSPrompter [12]	R-50	63.90	80.10	78.40	50.40	64.60	48.70	79.40	73.90	90.20	45.90	56.20	52.40	56.70	75.70	60.00	45.50
	YOLACT [109]	R-101	64.80	73.50	86.80	53.60	63.80	50.30	58.50	67.80	66.20	93.90	**75.00**	67.70	63.90	58.50	56.70	35.20
	SOLOv2 [111]	R-101	63.00	80.10	84.90	48.30	62.80	51.80	49.40	62.00	53.90	93.70	67.90	78.30	66.70	46.40	60.60	37.90
	MS RCNN [104]	R-101	70.90	85.60	85.70	57.40	66.60	57.50	59.90	82.70	73.80	93.80	74.40	76.70	**76.50**	66.90	59.50	46.20
	Instaboost [112]	R-101	69.70	87.20	84.40	55.80	62.50	56.10	57.20	84.10	74.00	93.00	69.50	78.00	72.90	64.00	55.60	51.50
Contour-based	E2EC [32]	DLA-34	67.10	86.37	84.59	**65.12**	65.03	**84.22**	85.32	83.39	72.97	92.99	67.40	74.59	62.10	38.88	58.89	44.04
	PolySnake [36]	R-50	66.60	87.64	82.75	56.49	68.03	55.73	82.92	74.12	94.14	67.54	66.54	31.39	69.64	74.61	58.49	29.62
	UniconDet (Ours)	R-50	74.70	93.83	86.28	47.82	65.08	71.37	**88.38**	89.59	94.85	79.69	74.24	51.61	68.90	78.92	63.52	66.19
		R-101	**75.90**	93.99	87.42	49.34	68.44	72.95	88.00	89.34	**95.48**	79.13	74.33	57.73	66.50	79.96	**66.21**	**69.16**
		R-152	75.38	**94.56**	**87.70**	49.28	**72.60**	73.56	88.33	89.00	95.34	78.01	72.87	58.74	67.72	**80.61**	61.44	60.92

multigrained object detection datasets, i.e., MG-DOTA and MG-NV10 are employed for evaluation. At the same time, 12 famous detectors involving several InSeg detectors (e.g., MaskRCNN [103], MS RCNN [104], QueryInst [105], YOLACT [109], PANet [62], RSPrompter [12], CATNet [9], Mask2Former [15], SOLOv2 [111], and Instaboost [112]) and contour-based methods (e.g., E2EC [32] and PolySnake [36]) are chosen as the comparisons. Next, the detailed analysis will be discussed as follows.

First, regarding the self-built dataset named MG-DOTA, the comparison results are reported in Table VII. From Table VII, we can find that our proposed UniconDet can achieve the best performance of 74.70%, 75.90%, or 75.38% mAP with various backbone equipment (i.e., R-50, R-101, or R-152). It can surpass two-stage pixel-based methods (i.e., MaskRCNN [103], QueryInst [105], MS RCNN [104], and Instaboost [112]) by around 6.6%, 7.9%, 5.0%, and 6.2% mAP, as shown in the second, fourth, ninth, and tenth rows of Table VII. In addition, compared with the contour-based methods, our proposed UniconDet can exceed E2EC [32] by 8.8% mAP and PolySnake [36] by 9.3% mAP, and the gains can be found in the 11th and 12th rows of Table VII. These performance gains indicate that the proposed UniconDet has better multigrained object detection ability than currently existing pixel-level and contour-based methods. Specifically, in order to further analyze the performance gains of different categories, the AP results of each category would be discussed in detail. As the results reported in the 5th∼19th columns of Table VII, the proposed UniconDet can achieve the best performance in the categories of SBF, SP, and H (i.e., labeled by yellow color) with HBB detection requirement. Remarkably, for small-scale objects of H, the proposed UniconDet can significantly surpass the other comparisons by at least 13% AP. Subsequently, regarding OBB detection requirement (i.e., highlighted by green color), our proposed UniconDet can also provide the best AP on some challenging categories of SV and S, which have the small-scale and multiscale characteristics, respectively. Meanwhile, it can achieve acceptable results in the categories

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TABLE VIII
COMPARISON EXPERIMENTS ABOUT MULTIGRAINED DETECTION TASK ON THE MG-NV10. * RED: INSEG; GREEN: OBB; AND YELLOW: HBB

	Method	Backbone	mAP	AL	BD	S	TC	BC	HB	B	V	GTF	ST
Pixel-based	PANet [62]	R-50	91.20	96.80	**97.00**	87.30	91.00	74.30	91.00	86.60	91.10	98.30	98.20
	MaskRCNN [103]	R-101	91.80	95.20	96.60	88.70	89.20	83.20	89.00	84.80	93.80	99.20	**98.30**
	MS RCNN [104]	R-101	92.68	**98.80**	96.50	89.10	89.80	83.40	90.70	86.90	**93.90**	**99.90**	97.80
	QueryInst [105]	R-101	72.40	86.20	85.40	69.50	55.70	64.80	65.10	65.20	72.10	99.90	60.30
	YOLACT [109]	R-101	78.10	54.90	96.30	80.20	49.40	77.00	81.40	77.70	87.70	97.80	78.10
	RSPrompter [62]	R-101	91.20	96.80	97.00	87.30	91.00	74.30	91.00	86.60	91.10	98.30	98.20
	CATNet [62]	R-101	93.40	98.70	91.20	**99.20**	95.40	87.20	**97.00**	99.90	85.70	86.80	92.50
Contour-based	E2EC [32]	R-101	88.00	93.99	95.22	88.88	90.30	73.98	84.59	77.91	84.20	94.74	96.32
	PolySnake [62]	DLA-34	90.60	97.49	87.63	90.86	**97.78**	89.36	93.47	97.87	87.99	78.39	84.86
	UniconDet (Ours)	R-50	94.03	91.89	94.30	98.58	96.90	92.24	96.08	**99.93**	83.83	89.26	97.21
		R-101	94.19	94.82	95.76	98.80	96.02	**92.68**	96.49	99.88	87.51	84.62	95.35
		R-152	**94.23**	92.91	93.56	98.26	96.83	92.20	96.30	99.91	89.78	86.76	96.06

of B, LV, and BC. Next, about the InSeg detection requirement (i.e., highlighted by red color), our proposed UniconDet can perform the best results on the categories of P, BD, and R. Summarily, when different detection requirements of HBB, OBB, or InSeg are integrated into one dataset, our proposed UniconDet can provide superior object detection performance and tackle well arbitrary-shaped object detection from complex RSI.

Additionally, the visualization results of our proposed UniconDet and famous detectors (i.e., RSPrompter [12], CATNet [9], and QueryInst [105]) are presented in Fig. 7. Specifically, as shown in Fig. 7, the column (c) of our proposed UniconDet can provide more accurate and stable detection performance than the other comparisons. In detail, in the first row of Fig. 7, under a complex inshore port scene, there are a lot of co-scene categories of objects with vast scale variations. Facing this challenging situation, more small-scale Ss and large-scale HBs can be accurately detected by our proposed UniconDet, as shown in the red rectangles of column (c). However, other comparison results of RSPrompter [12], CATNet [9], and QueryInst [105] are shown in columns (d)–(f), which all appear with varying degrees of missed detections. Then, in the second row of Fig. 7, under a very low contrast oceanic scene, there are some Ss that are hard to distinguish from the sea surface, but as shown in column (c), our proposed UniconDet can well detect these Ss compared with columns (d)–(f). Next, the visual results for small-scale AL detection with relatively complicated shapes are shown in the third and fourth rows of Fig. 7. Compared with the columns (d)–(f), our proposed UniconDet not only can detect the Hs from the background but also provides better and integral contour detection performance for ALs. Finally, in the fifth and sixth rows of Fig. 7, we can see that our proposed UniconDet can also perform well in multigrained object detection under typical urban scenes. As shown in the red rectangles of column (c), several SVs can be effectively detected from the shade of trees, which indicates the robustness of our proposed UniconDet for resisting background interference. Furthermore, it can be found that the proposed UniconDet exhibited better integrity in object detection, even when overlapping occurs among different objects. Otherwise, the pixel-based methods of RSPrompter [12], CATNet [9], and QueryInst [105] would easily produce fragmentary predictions for object edges. This is because of the pixelwise prediction overlooking the global and integral shape contour of objects, as presented in the red rectangles of columns (d)–(f).

Second, to further prove the robustness of the proposed UniconDet for the multigrained object detection, the self-built dataset called MG-NV10 is also employed for evaluation, and the results are reported in Table VIII. As shown in the last row, our proposed UniconDet still surpasses other detectors while achieving the SOTA result of 94.23% mAP on the MG-NV10 dataset. According to the above results, it has been proven that the proposed UniconDet is capable of detecting objects in any granularity simultaneously, and it is hopeful to make contributions to the development of remote sensing object detection in practical applications. Apart from the analysis of the proposed UniconDet, as shown in the second and third rows of Table VIII, these two-stage detectors such as MaskRCNN [103] and MS RCNN [104] can still maintain pretty overall performances of 91.80% and 92.68% mAP on the MG-NV10. In contrast, several one-stage detectors such as QueryInst [105], YOLACT [109], and E2EC [32] obtain bad overall performances that are lower than 90.00% mAP. Accordingly, it can be concluded that the model design of the two-stage paradigm can provide a more reliable performance guarantee for multigrained object detection benchmarks. Furthermore, as reported in the sixth and seventh rows of Table VIII, RSPrompter [12] achieves an overall performance of 91.2% mAP through transferring the powerful generalization ability of the large-scale foundation SAM model into the remote sensing scene. CATNet [9] was proposed to improve the feature extraction process by addressing the severe scale variations, very low contrasts, and clustered distributions in the RSI. As a matter of course, CATNet [9] can also achieve a competitive result, i.e., 93.40% mAP.

C. Ablation Analysis

In this section, we provide detailed experiments and analysis about the designed FCPM, MVCA, and C^2-ERL within the

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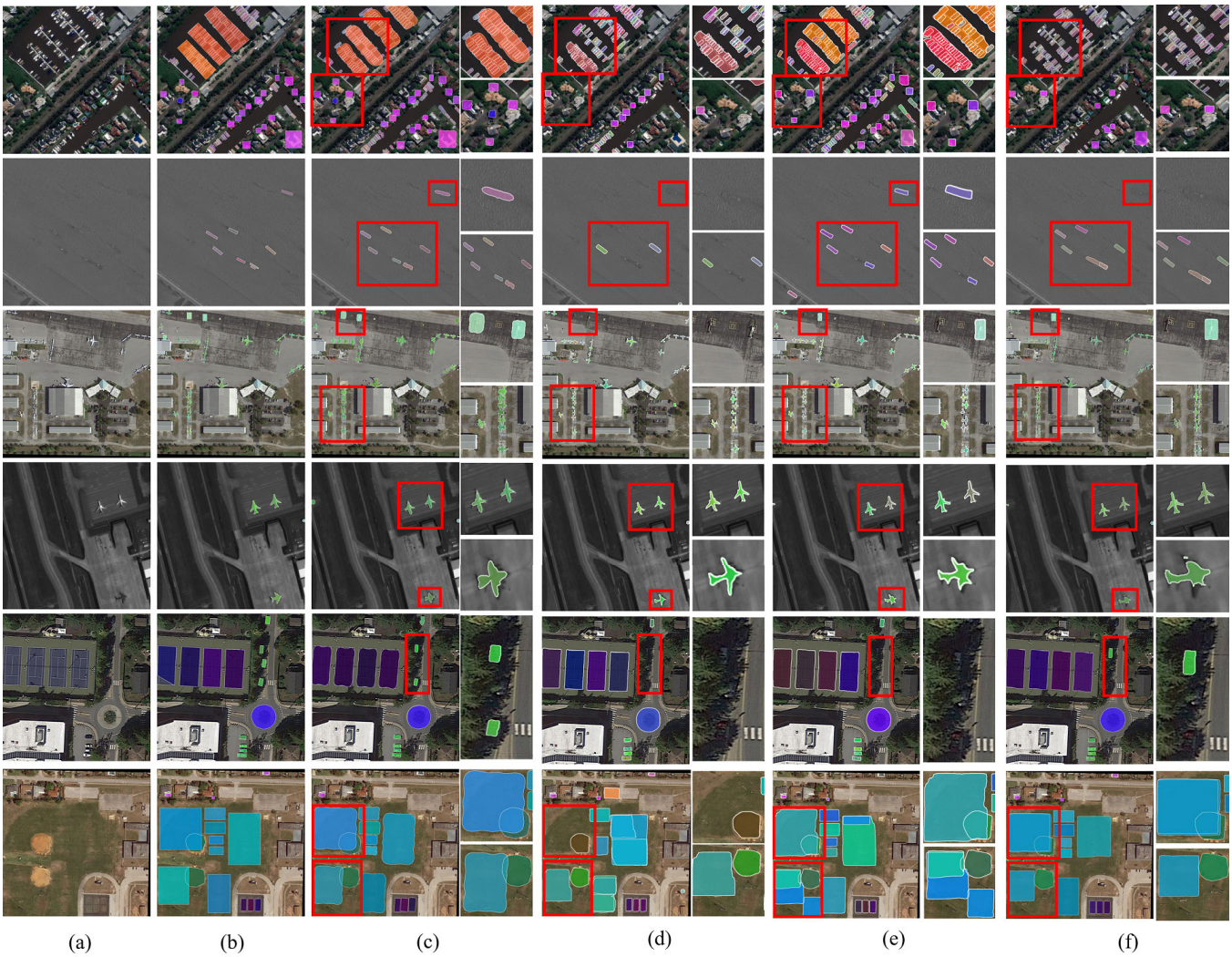


Fig. 7. Qualitative comparison of the proposed UniconDet and classic methods on multigrained object detection. (a) Original image. (b) Ground truths. (c) Results of UniconDet. (d) Results detected by RSPrompter [12]. (e) Results of CATNet [9]. (f) Results of QueryInst [105].

TABLE IX
ABLATION EXPERIMENTS ON MULTIGRAINED DOTA

Method	Module				mAP(%)
	Contour Point	FCPM	MVCA	C^2-ERL	
Baseline	✓				65.54
		✓			66.81
		✓	✓		67.88
UniconDet		✓	✓	✓	**68.97**

TABLE X
DISCUSSION ON THE CONTOUR GEOMETRIC MODELING

The Methods of Contour Geometric Modeling			mAP(%)
Fourier degree	Point Offset		60.91
	Polar Radius	**w/o.** Norm.	65.45
		w. Norm.	**66.81**
Chebyshev polynomials	Polar Radius	**w.** Norm.	62.00

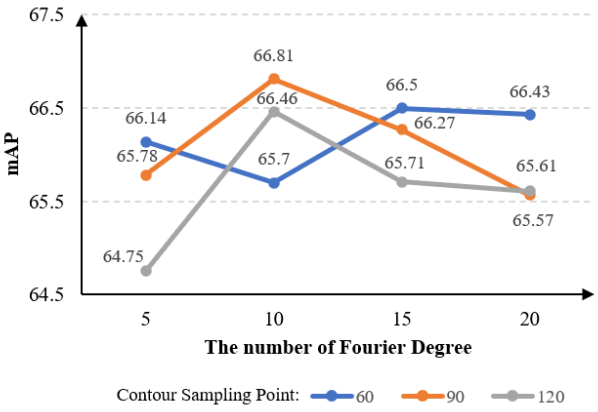


Fig. 8. Performance change of FCPM modeling method with different sample points and Fourier degree.

proposed UniconDet, based on the MG-DOTA dataset. As a premise, CenterNet [49] with R-50 is adopted as the baseline detector. Next, as reported in Table IX, for clearly observing the effectiveness of the designed parametric modeling strategy and specific modules, they would be successively added into the baseline detector for analysis.

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TABLE XI
DISCUSSION ABOUT THE NUMBER OF TRANSFORMED VIEWS

The number of transformed views	1	2	3	4
mAP(%)	66.52	67.88	**67.89**	66.33
Params (M)	28.52	28.97	29.42	**29.87**

TABLE XII
ABLATION EXPERIMENTS ON THE SUBMODULE OF C^2-ERL

Module	Region-Guiding Center Prediciton		Cascade Contour Propagation	mAP(%)
	w/ Gradient Truncation	**w/o** Gradient Truncation		
Baseline + FCPM + MVCA				67.88
	✓			68.79 (↑ 0.91%)
		✓		67.47 (↓ 0.41%)
			✓	68.46 (↑ 0.58%)
	✓		✓	68.97 (↑ 1.09%)

First, as reported in the first row of Table IX, the baseline detector can obtain 65.54% mAP by adopting the modeling way of spatial discrete contour points. Subsequently, upon selecting the designed FCPM, as illustrated in the second row of Table IX, the overall performance can be increased by 1.27% mAP, which means that the designed FCPM can effectively promote more accurate object contour detection. Furthermore, from the third row of Table IX, after further incorporating the designed MVCA, it can achieve a 1.07% mAP improvement, which indicates that the designed MVCA has the ability to assist the model in understanding various object contours by exploring the coupled relations between transformed views. Lastly, when the designed C^2-ERL is also introduced, it can be found that the C^2-ERL can bring a 1.09% mAP gain, and the combination of three proposed modules totally brings a 3.43% mAP improvement. In general, the designed parametric modeling strategy of FCPM, feature extraction way of MVCA, and regression layer of C^2-ERL are very effective for improving the performance of multigrained object detection. Next, the detailed ablation analysis would be elaborated as follows.

1) Ablation Analysis on FCPM: As illustrated in Section III-A, the settings of Fourier degrees and discrete contour sampling points are two critical factors for FCPM. Consequently, these two factors are discussed through ablation experiments. Specifically, as shown in Fig. 8, there are several change curves of mAP under different settings of Fourier degrees and discrete contour sampling points. From the initial points, it can be seen that when the number of Fourier degrees is set as a minimum number, adopting more sampling points would present worse performance. This indicates that insufficient Fourier degrees cannot support describing fine-grained contours with too many sampling points. Then, according to each curve shown in Fig. 8, when the number of sampling points is set to 60, as drawn in the blue curve, the mAP presents a trend of growth though it has a fluctuation under the situation where the Fourier degree is set to 10. Next, as the number of sampling points increases to 90 and 120, such as the orange and gray curves, they both reach the peak value of mAP when the Fourier degree is equal to 10. After that, the orange and gray curves indicate that mAP begins to decrease when the Fourier degree is increasing. The reason is analyzed to be that excessive Fourier degrees would bring the burden of parameter regression meanwhile lead to performance degradation. In general, when the Fourier degree is set to 10, the number of sampling points is chosen to be 90. The proposed UniconDet can achieve the best result of 66.81% mAP. So that, subsequent experiments are based on these hyperparameters.

Apart from the discussion of hyperparameters for FCPM, different contour parametric modeling ways are further discussed. First, different processing methods of contour points before the Fourier transform are analyzed in detail. As shown in the first row of Table X, the point offset is adopted to replace the polar radius in the polar coordinate system. Specifically, the point offset is calculated by subtracting the coordinate of the center point (x_{ct}, y_{ct}) from the coordinate of the contour point $(x_n, y_n), n \in [1, N]$ in the Cartesian coordinate system. And then, the calculated set of $(\Delta x_n, \Delta y_n), n \in [1, N]$ is used for Fourier transform. Then, we can see that it achieves the worst performance, i.e., 60.91% mAP in the first row of Table X. The reason is that the point offset usually presents positive or negative values, which makes it more challenging to reconstruct these offsets by inverse Fourier transform than the polar radius with positive values. Besides, in order to verify the necessity of polar radius normalization for contour points, the ablation discussion is shown in the second and third rows of Table X. By comparison, the situation of *w. Norm.* exceeds the *w/o. Norm.* by 1.36% mAP. This demonstrates that scale-invariance normalization is very effective for modeling remote sensing objects with vast scale variations. Meanwhile, another global contour modeling approach of Chebyshev polynomials is also introduced as a comparison. As reported in the last row of Table X, when the normalized polar radius is transformed into Chebyshev polynomials for parameter regression, it achieves 62.00% mAP. In general, the designed FCPM can be proven to be a promising parametric modeling method for achieving arbitrary-shaped contour detection.

2) Ablation Analysis on MVCA: Theoretically, the number of transformed views involved in the designed MVCA should correlate positively with the performance because more transformed views should assist the model in perceiving a greater variety of object contour distributions. Nevertheless, adopting more transformed views would inevitably lead to an increase in the number of parameters, i.e., Params (M). To this end, the influence of the number of transformed views in the designed MVCA is discussed. As the results reported in Table XI, the mAP and Params (M) are calculated by setting different transformed views. Then, we can find that when scaling up the number of transformed views to 3, the mAP constantly grows. Next, when the number of transformed views changes from 3 to 4, the performance of mAP would be declined by 1.56%. Conversely, the Params (M) steadily increases as the number of transformed views increases. We analyze the reason

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TABLE XIII
COMPARISON ANALYSIS ABOUT MODEL COMPLEXITY AND INFERENCE EFFICIENCY

Method	Image Size	Backbone	DIOR (HBB@mAP)	DOTA (OBB@mAP)	iSAID (InSeg@mAP)	Params(M)	GFLOPs	FPS
RT-DETRv2 [114]	640×640	R-18	77.10	-	-	21.89	59.94	10.14
E2EC [32]	800×800	DLA-34	-	-	60.20	29.77	60.14	15.10
PolySnake [36]	800×800	DLA-34	-	-	61.70	22.04	66.07	11.40
RSPrompter [12]	1024×1024	R-50	58.50	-	63.50	113.67	620.54	11.23
MaskRCNN [103]	800×800	R-50	-	-	62.10	43.82	186.21	20.08
Instaboost [112]	800×800	R-50	-	-	56.90	95.84	365.06	13.94
CATNet [9]	832×832	R-50	-	-	63.60	56.68	261.12	15.25
ReDet [56]	800×800	ReR-50	-	80.10	-	31.60	59.81	13.96
YOLOv8 [70]	640×640	CSPD-53	74.30	-	-	11.14	22.38	**102.04**
MS RCNN [104]	800×800	R-101	-	-	62.00	78.73	236.10	15.22
MSFCNet [2]	800×800	RS-101	70.63	-	-	46.80	126.90	24.23
STD-O [86]	800×800	ViT-B	-	**81.66**	-	**122.45**	**1586.78**	4.09
SOCDet [102]	1024×1024	SOCDB	-	78.83	-	9.98	16.06	26.57
UniconDet (Ours)	800×800	R-18	73.64	77.24	64.20	17.58	135.63	41.18
	800×800	R-50	75.80	78.40	**66.60**	28.97	159.54	27.83
	1024×1024	R-101	**77.30**	78.45	66.30	48.86	229.65	26.31

and find that redundant information is present among those transformed views. Besides, too many parameters are difficult to be optimized within the same training period. Consequently, the number of transformed views is set to 2, aiming to achieve a tradeoff between mAP and Params (M).

In addition, to intuitively observe the impact of the designed MVCA on contour parameter perception, some visualizations of predicted contour periodic distributions are provided in Fig. 9. Notably, three typical remote sensing categories, including Ss, ALs, and TCs, are selected as examples. In more details, the second row denotes the ground truth of object contour distributions. The third row denotes predicted contour distributions from the UniconDet *w/o.* MVCA. The fourth row denotes predicted contour distributions from the UniconDet *w.* MVCA. Regarding the second row as a reference, it can be found from the gray elliptical that the results of the UniconDet *w.* MVCA exhibit more delicate contour distributions than those of the UniconDet *w/o.* MVCA for different kinds of objects. Specifically, as shown in Fig. 9(b), with respect to the contour of the AL, it is obvious that the UniconDet *w.* MVCA can predict more complicated distribution changes than those of the UniconDet *w/o.* MVCA. At the same time, as for the S and TC shown in Fig. 9(a) and (c), the designed MVCA can assist the UniconDet in fitting the object contours more accurately. Therefore, these visualizations demonstrate that the designed MVCA can set up an effective perception ability for contour parameter learning, which relies on exploring the coupled relations among different contour sampling periodic distributions from transformed views.

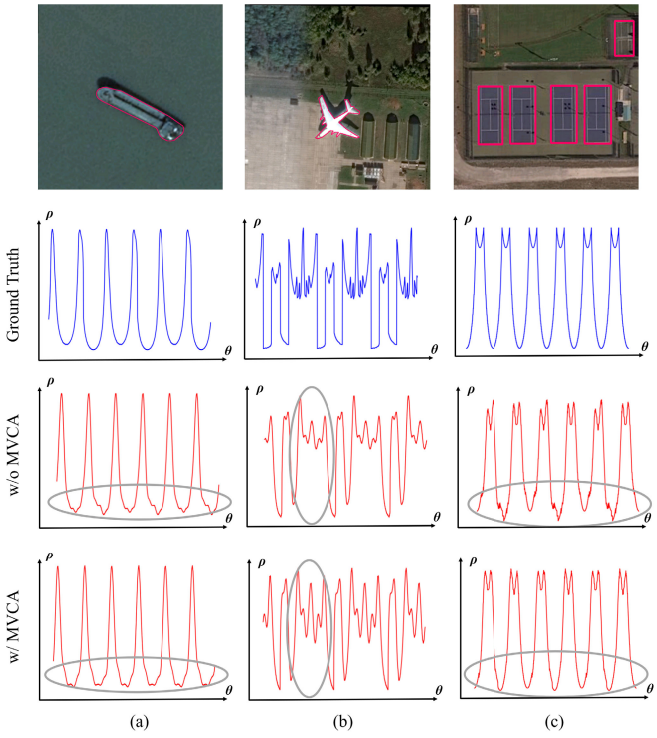


Fig. 9. Visualizations about the effects of MVCA. (First row) Original images with object contours. (Second row) Real contour periodic distribution. (Third row) Contour periodic distribution predicted from the UniconDet *w/o.* MVCA. (Fourth row) Contour periodic distribution predicted from the UniconDet *w.* MVCA. (a) Denotes an example of a ship with OBB. (b) Indicates an example of an airplane with InSeg. (c) Denotes an example of tennis courts with HBB.

3) Ablation Analysis on C²-ERL: To advance the prediction of Fourier contour parameters, the region-guiding center prediction and cascade contour propagation have been designed into the C²-ERL module. First, to quantitatively analyze their performance, the gains brought by them are reported in Table XII. As shown in the second and third rows, the region-guiding center prediction can bring a 0.91% mAP improvement when truncating the gradient backpropagation from the center point prediction to region-level instance prediction. In contrast, the region-guiding center prediction *w/o.* gradient truncation leads to a performance decrease of 0.41% mAP. The reason may be that the weight parameters updated by hybrid supervised tasks have conflicts on the

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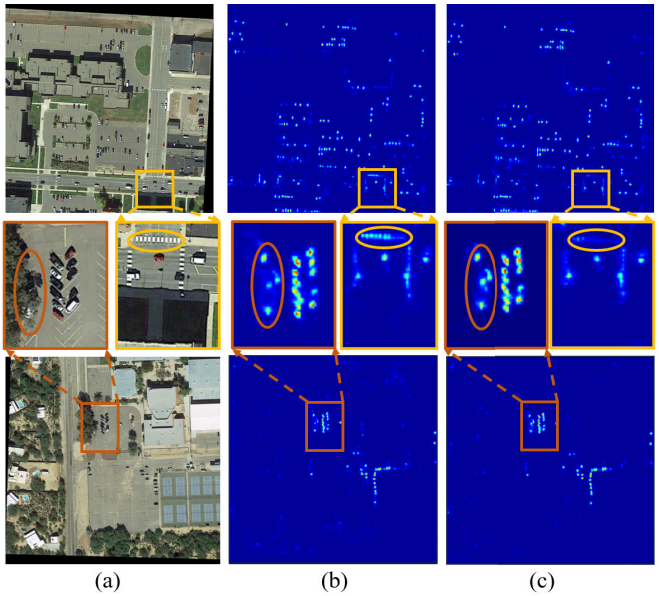


Fig. 10. Comparison of the results of center points. (a) Original image. (b) Heatmap of center points from UniconDet *w/o.* C^2-ERL. (c) Heatmap of center points from UniconDet *w.* C^2-ERL.

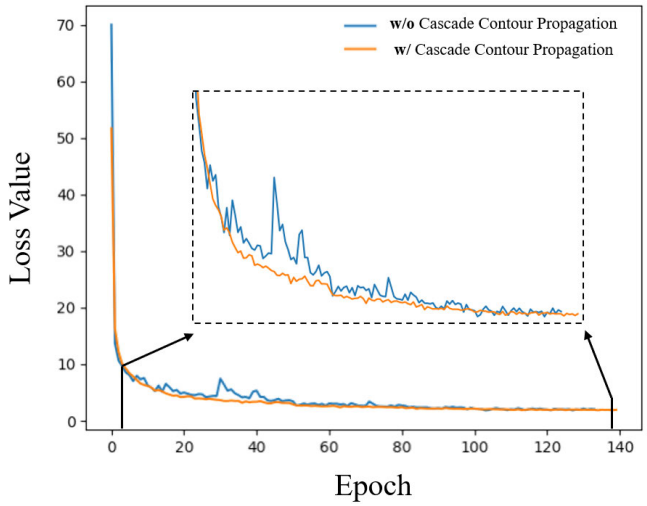


Fig. 11. Change curves of training loss under the proposed UniconDet *w/o.* or *w.* cascade contour propagation.

region-level instance and pixel-level center point predictions. In addition to guiding the prediction of center points from the feature level, during the inference phase, the heatmap of the predicted region-level instance mask is used to multiply with the heatmap of center points. As a result, the quality of the region-level instance mask is essential, and it is necessary to truncate the gradient to avoid the impact of other branches on its results. Subsequently, from the fourth row of Table XII, it can be found that the stand-alone cascade contour propagation can also achieve a 0.58% mAP gain. At the end, when combining the region-guiding center prediction with the cascade contour propagation, a 1.09% mAP gain can be obtained, which is shown in the fifth row of Table XII.

From another view, the effect of C^2-ERL would be qualitatively illustrated in the following. As shown in Fig. 10, the predicted results of center points are displayed in the form of a heatmap, where the warmer color represents a higher confidence score. In detail, by comparing Fig. 10(b) and (c), the C^2-ERL has the capability to suppress FP objects. Specifically, it can be observed that these lane marking lines (e.g., zebra-stripe) can be filtered in Fig. 10(c) even though they have a similar shape and size to the SVs. In addition, these Vs that are covered by trees have a higher confidence score in Fig. 10(c) than in Fig. 10(b). By the comparison, it can be concluded that the guidance from region-level prediction can provide sufficient information description for center point prediction. Moreover, regarding the effectiveness of cascade contour propagation, we analyze the effect brought by it from the perspective of training loss. As shown in Fig. 11, due to the prior information of the previous contour regression layer, we can find that the training loss has a lower initial value and faster loss decrease. This indicates that the cascade contour propagation not only can bring performance gain, but also is beneficial for model convergence.

4) Model Complexity and Efficiency Analysis: In addition to accuracy, the model complexity and inference efficiency are also important criteria for evaluating the detectors. Therefore, as illustrated in Table XIII, 13 famous objectors designed for HBB, OBB, and InSeg tasks are introduced for comparison. In practice, the ten same images are prepared for each detector to evaluate the number of parameters [Params (M)], giga floating-point operations (GFLOPs), and frames per second (FPS). As shown in the 12th row of Table XIII, it can be found that the STD-O [86] has achieved the SOTA performance, i.e., 81.66% mAP in the OBB task of the DOTA dataset [51]. However, with respect to model complexity, STD-O [86] has the largest Params (M) and highest GFLOPs, i.e., 122.45 Params (M) and 1586.78 GFLOPs, which indicates that STD-O [86] possesses the most complexity of model design. Meanwhile, because of its heavy model design, STD-O [86] also performs the worst in inference efficiency, i.e., 4.09 FPS. In addition, RSPrompter [12] is designed by combining with the advanced large-scale pretrained model of SAM [113], but it also performs a poor outcome in terms of model complexity, as reported in the fourth row of Table XIII. It can be concluded that although advanced pretraining approaches and powerful foundation models can facilitate the performance improvement in the remote sensing domain, they also significantly increase the complexity of the models and reduce their inference efficiency, which is quite unfriendly to low-resource environments. As reported in the sixth and seventh rows of Table XIII, when employing smaller image sizes and shallower backbones, these conventional two-stage detectors still exhibit heavier Params (M) and greater GFLOPs than our proposed UniconDet. Besides, some SOTA detectors that focus on real-time or resource-efficient applications are also introduced for comparison. As reported in the 1st, 9th, and 13th rows of Table XIII, RT-DETRv2 [114], YOLOv8 [70], and SOCDet [102] obviously achieve significant advantages in terms of Params (M) and GFLOPs. By comparing as a whole, although our proposed UniconDet is at a moderate level with respect to GFLOPs, the third-lowest Params (M) was achieved

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when equipped with the backbone of R-18. Furthermore, with respect to the comparison of FPS, the famous real-time detector called YOLOv8 [70] obtains the highest inference efficiency; meanwhile, our proposed UniconDet also achieves the second-highest FPS. To sum up, by comparing with the various SOTA detectors focusing on accuracy and efficiency, one conclusion is that with the unified FCPM parametric modeling, MVCA, and C^2-ERL modules, our proposed UniconDet maintains an acceptable model complexity and inference efficiency while ensuring remarkable performance across different tasks. This fully proves that the proposed UniconDet is very robust due to its universal architecture design.

V. Conclusion

In this work, a new unified remote sensing object framework is presented to integrate various object detection requirements (i.e., HBB, OBB, and InSeg) into a more challenging task for arbitrary-shaped contour detection. Naturally, a multigrained object detection benchmark is also set up to make the research efforts concentrate together and facilitate the development of remote sensing object detection toward a unified solution. At the same time, a novel parametric modeling strategy of FCPM, feature extraction way of MVCA, and regression layer of C^2-ERL are designed into a unified structure called UniconDet, which can project arbitrary-shaped object contours from the spatial domain into the frequency domain for effective model learning, while achieving SOTA performance in multigrained object detection. Then, extensive experiments are carried out on benchmarks of HBB, OBB, InSeg, and the newly built task of multigrained object detection. The results indicate that the proposed UniconDet can achieve SOTA performance in each benchmark only relying on a unified model and unique detection head, which demonstrates that the proposed UniconDet can effectively cope with various detection requirements of different category objects from complex remote sensing scenes and be easily applied for practical object detection. According to the experimental result analysis, in future work, the objects with extreme nonconvex contours need to be further studied for improving their detection performance. Meanwhile, this study will also be devoted to setting up a unified detector that adapts well to multisource and multitask detection.

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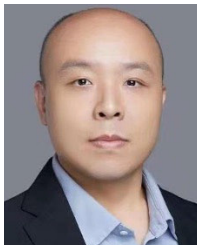
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